Robust Structural Estimation under Misspecified Latent-State Dynamics*

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Abstract

Estimation and counterfactual analysis in dynamic structural models rely on assumptions about the dynamic process of latent variables, which may be misspecified. We propose a framework to quantify the sensitivity of scalar parameters of interest (e.g., welfare, elasticity) to such assumptions. We derive bounds on the scalar parameter by perturbing a reference dynamic process, while imposing a stationarity condition for time-homogeneous models or a Markovian condition for time-inhomogeneous models. The bounds are the solutions to optimization problems, for which we derive a computationally tractable dual formulation. We establish consistency, convergence rate, and asymptotic distribution for the estimator of the bounds. We demonstrate the approach with an infinite-horizon dynamic demand model for new cars in the United Kingdom, Germany, and France. Under the perturbed dynamics, price elasticities deviate by at most 4.94% from the reference estimates, yet consumer surplus from an electric vehicle subsidy varies by up to 85.61%.

Keywords: Dynamic structural models, Sensitivity analysis, Misspecified dynamics

JEL Codes: C14, C18, C51, C61

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1 Introduction

Dynamic structural models are useful tools for counterfactual policy analysis in various fields of economics. The dynamic process of latent variables is a key feature of these models, as it captures the persistence of unobserved factors that affect agents' decisions over time. Examples of potentially serially correlated latent variables include product characteristics in demand estimation (Nair, 2007; Schiraldi, 2011; Gowrisankaran and Rysman, 2012), search costs in consumer search (Koulayev, 2014), firm productivity in trade (Piveteau, 2021), patent profitability in optimal stopping (Pakes, 1986), quality in technology adoption (De Groote and Verboven, 2019), health shocks in insurance (Fang and Kung, 2021), and beliefs about ability in labor economics (Miller, 1984; Arcidiacono et al., 2025).

Assumptions about the dynamic process governing the serial dependence of latent variables are central to the estimation and counterfactual analysis in dynamic structural models. These assumptions capture agents' uncertainty about the future, such as a consumer's uncertainty about future product characteristics in demand estimation. The misspecification of these assumptions can lead to biased estimates of future continuation values, which in turn bias counterfactual predictions (e.g., welfare and elasticity). However, the direction and magnitude of this bias are unclear, because the models are dynamic and nonlinear. This raises the need for sensitivity analysis of empirical results to these distributional assumptions.

In this paper, we propose a framework to quantify this sensitivity by perturbing a reference dynamic process to compute bounds on a scalar parameter of interest. The scalar parameter (e.g., welfare and elasticity) is a function of model primitives, such as model parameters, the transition distribution of latent variables, and the value function in dynamic structural models. In our proposed framework, the bounds on this scalar parameter are the solutions to constrained optimization problems whose feasible region (identified set) is defined by moment conditions for estimation, structural constraints (e.g., the Bellman equation), and the perturbation set around the reference transition distribution. The distribution that achieves the bound is called the worst-case distribution.

A central challenge is to define the perturbation set in a way that simplifies computation while maintaining key structural features of the dynamic process of latent variables. For time-homogeneous models, the structural feature is the stationarity condition, i.e., the perturbed dynamic process must be stationary. For finite-horizon, time-inhomogeneous models, the perturbed trajectory of latent variables must be Markovian. In addition, the terminal distribution is fixed because it can be nonparametrically point identified (Lewbel, 2000; Matzkin, 2007). Because the stationarity and Markov conditions are functional constraints imposed directly on the distribution being optimized, they are computationally difficult to impose. We contribute to the sensitivity analysis literature (e.g., Christensen and Connault, 2023) by providing a computationally tractable framework to deal with these constraints.

There are further practical challenges. First, the properties (e.g., closed-form, smoothness) of the worst-case distribution are typically unclear. Second, the expectations that define the scalar parameter, the model-implied moments, and the structural constraints are all calculated with respect to the perturbed distribution. Because this distribution is itself an optimization variable that changes during the optimization, the numerical integration can be difficult to implement. Third, the value function in dynamic structural models is infinite-dimensional due to the serial dependence of latent variables, which complicates the optimization problem further.

To address these challenges, we define the perturbation set as a Kullback–Leibler (KL) divergence ball around the reference distribution. For the time-homogeneous case, the reference distribution is a joint distribution of current and future latent variables. For the time-inhomogeneous case, it is the joint distribution of the entire trajectory of latent variables. The KL radius controls the size of the perturbation set. We then employ the Optimal Transport (OT) framework to impose structural constraints on the distribution of latent variables. In the time-homogeneous case, the stationarity condition requires that the marginal distributions of current and future latent variables of the perturbed joint distribution coincide, which can be imposed using OT. This marginal constraint, together with the KL divergence penalty from the problem's Lagrangian, yields a computationally tractable Entropic Optimal Transport (EOT) problem. In the time-inhomogeneous case, the perturbed trajectory must be Markovian, and the terminal distribution is fixed. We also show how to formulate this as an EOT problem.

The EOT problem can be solved using its dual formulation, which has three key advantages over the primal formulation. First, it provides a closed-form expression for the worst-case distribution and characterizes its smoothness. During our proposed optimization algorithm, this closed-form expression is used to update the value function in dynamic structural models. Second, the Sinkhorn algorithm (Sinkhorn and Knopp, 1967; Cuturi, 2013) allows us to solve the EOT problem and compute the worst-case distribution efficiently. Finally, because the expectations in the dual problem are taken with respect to the reference distribution, an appropriate numerical integration method can be chosen in advance.

Then, we consider three complementary sensitivity measures to interpret the results. First, the *global sensitivity* approach computes the largest deviation from the reference value. It increases the KL radius until a pre-specified maximum is reached. We show that it provides a tractable approximation to the nonparametric bounds on the scalar parameter when the KL divergence constraint is removed. Moreover, we derive an explicit upper bound on the approximation error, which enables us to control the error within a desired level. Second, the *local sensitivity* approach analyzes the effect of small perturbations. It computes the right derivatives of the bounds with respect to the KL radius, which serves as our local sensitivity measure. Finally, the *robustness metric* approach, inspired by Spini (2024), computes the smallest deviation from the reference distribution required to produce sensitive results. It is the smallest KL divergence from the reference distribution required for the scalar parameter's value to fall below a user-specified threshold (e.g., 5% below the reference value). In addition to our three sensitivity measures, we can also estimate an alternative model and set the radius as the KL divergence from the alternative to reference models.

For large sample properties, we propose an estimator for the bound, establishing its consistency and convergence rate. To this end, we first establish the consistency and convergence rate of the estimator of the identified set, following Chernozhukov et al. (2007). We then derive the asymptotic distribution of the plug-in estimator by proving the Hadamard directional differentiability of the bound.

We apply our framework to an infinite-horizon dynamic demand model for new cars in the UK, Germany, and France. We consider the sensitivity of the average industrywide price elasticity and consumer surplus (CS) in 2023 from an additional $3,000 electric vehicle subsidy. In the model, the indirect utility is a latent variable due to unobserved product characteristics, and its transition is typically modeled as an AR(1) process (e.g., Schiraldi, 2011; Gowrisankaran and Rysman, 2012). For the average price elasticity, we find that it is relatively robust to the distributional assumption, with global sensitivity at most 4.94% in the UK, 3.54% in Germany, and 2.60% in France. For CS, misspecification matters more: global sensitivity reaches 85.61% in Germany, 82.27% in the UK, and 51.52% in France.

1.1 Related Literature

Identification and estimation. Many papers have focused on identification in a range of dynamic structural models including finite mixture models (Kasahara and Shimotsu, 2009; Luo et al., 2022; Higgins and Jochmans, 2023, 2025), unobservable Markov processes (Hu and Shum, 2012), and counterfactual conditional choice probabilities in dynamic binary choice models (Norets and Tang, 2014). Berry and Compiani (2023) uses the generalized instrumental variable approach for unobserved state variables in dynamic discrete choice (DDC) models. In fixed effects DDC models, Aguirregabiria et al. (2021) considers identification of structural parameters using sufficient statistics, while Aguirregabiria and Carro (2024) studies identification of average marginal effects. Hwang (2024) employs proxy variables for the latent variables. Kalouptsidi et al. (2021c) proposes the Euler Equations in Conditional Choice Probabilities (ECCP) estimator. By leveraging finite-dependence properties and cross-sectional data, it identifies structural parameters in the presence of serially correlated market-level unobserved variables without distributional assumptions. Arcidiacono and Miller (2011) adapts the Expectation-Maximization algorithm to estimate DDC models with discrete latent types. Norets (2009) extends the Bayesian estimation of DDC models by Imai et al. (2009) to allow for serially correlated latent variables, while Blevins (2016) proposes a sequential Monte Carlo method. Chiong et al. (2016) shows identification in DDC models is an optimal transport problem under serial independence of utility shocks. In addition, Chen et al. (2011), Schennach (2014), and Fan et al. (2023, 2025) consider inference of finite-dimensional parameters in the presence of an infinite-dimensional parameter, namely the distribution of latent variables.

Our framework contributes to this literature in three ways. First, we focus directly on a scalar parameter (e.g., welfare, elasticity) rather than on the identified set of model primitives. Second, to analyze sensitivity, we consider a perturbation set around the reference distribution rather than the set of all distributions. Third, we complement identification strategies that do not rely on distributional assumptions. For example, the ECCP estimator of Kalouptsidi et al. (2021c) can be used to point identify the utility parameters. Then, we can conduct sensitivity analysis of the scalar parameter of interest to the distributional assumptions about the transition of market-level unobserved variables.

Sensitivity analysis and robustness. In DDC models, Kalouptsidi et al. (2021a,b) relax common normalizations on the utility function. Bugni and Ura (2019) considers the local misspecification of the transition density of observable variables, assuming the transition density is correctly specified in the limit (as the sample size goes to infinity) in DDC models. Andrews et al. (2017) considers a setting in which the moments are locally misspecified under the reference distribution. Subsequent work Armstrong and Kolesár (2021) constructs near-optimal confidence intervals in such models. Kitamura et al. (2013) considers the robust estimation under moment restrictions. Bonhomme and Weidner (2022) perturbs the reference model and assumes the size of the perturbation shrinks to zero as the sample size goes to infinity. Chen et al. (2024) relaxes the rational expectation assumption. Gu and Russell (2024) considers the identification of scalar counterfactual parameters using optimal transport. Spini (2024) studies the robustness of policy effects to changes in the distribution of covariates. Armstrong (2025) provides a selective review of misspecification in econometrics. Most closely related is Christensen and Connault (2023), who conducts sensitivity analysis with respect to parametric assumptions about the distribution of latent variables in structural models. However, they focus on relaxing the marginal distribution assumption while maintaining serial independence.

We complement this literature by providing a computationally tractable framework for robust structural estimation under misspecified latent-state dynamics. Our main contribution is to handle the key structural constraints that make such problems difficult: the stationarity condition in time-homogeneous models and the Markov property together with the fixed terminal distribution in time-inhomogeneous models. These constraints are not addressed in Christensen and Connault (2023), who focus on relaxing marginal distributional assumptions while maintaining serial independence. In addition, by leveraging EOT duality, we obtain a closed-form expression for the worst-case distribution, which allows us to overcome the computational challenges created by the infinite-dimensional value function arising from serial dependence in dynamic structural models.

Distributionally robust optimization (DRO). The literature on DRO (Kuhn et al., 2019; Rahimian and Mehrotra, 2019; Blanchet et al., 2022; Gao and Kleywegt, 2023; Wang et al., 2021) usually studies the uncertainty due to limited observability of data, noisy measurements, or estimation errors.

Our work is distinct in that we study the misspecification due to assumptions about the serial dependence of latent variables—a problem of model specification rather than data limitation. Moreover, because our framework can be treated as a moment-constrained DRO problem, we employ the minimax theorem (see Fan 1953 and Ricceri and Simons 1998, Theorem 1.3) to exchange the order of the supremum over Lagrangian multipliers (for the moment conditions and structural constraints) and the infimum over distributions in the perturbation set. This exchange allows a direct application of duality results from the DRO literature, which simplifies our proof of the dual formulation significantly.

Applied work. Building on the seminal work on the estimation of DDC models (Rust, 1987; Hotz and Miller, 1993; Aguirregabiria and Mira, 2002, 2007; Pesendorfer and Schmidt-Dengler, 2008; Arcidiacono and Miller, 2011), most applied work assumes the serial independence of utility shocks. In the presence of serially correlated latent variables, parametric models are often used, as seen in the works of Wolpin (1984); Schiraldi (2011); Gowrisankaran and Rysman (2012); Blevins et al. (2018); Piveteau (2021) and others mentioned in the introduction.

We contribute to this literature by developing a computationally tractable framework for sensitivity analysis of scalar parameters of interest to these distributional assumptions. We demonstrate our framework through an empirical application to an infinite-horizon dynamic demand model for new cars in the UK, Germany, and France.

Outline: Sections 2 and 3 present our framework for time-homogeneous and time-inhomogeneous models. Section 4 establishes the large sample properties of our estimator. Section 5 introduces three sensitivity measures. Section 6 discusses practical implementation. Section 7 presents an empirical application. Section 8 concludes. Appendix A presents additional examples. All proofs are in Appendix B.

Notation: Let $U \in \mathcal{U} \subseteq \mathbb{R}^d$ be a vector of latent variables with support $\mathcal{U}$ where $\mathcal{U}$ is assumed to be Polish. Let $\mathcal{P}(\mathcal{U})$ be the space of Borel probability measures on $\mathcal{U}$ and $\mathcal{B}(\mathcal{U})$ be the Borel σ-algebra on $\mathcal{U}$. In this paper, all measures are assumed to be absolutely continuous with respect to the Lebesgue measure. Denote by $\mathbb{E}_F[\cdot]$, $\mathbb{E}_x[\cdot]$ the expectations with respect to $F \in \mathcal{P}(\mathcal{U})$ and the probability distribution of the random variable x, respectively. For variables in a stationary dynamic context, a prime (e.g., x') denotes the variable's value in the next period. Let $F_1, F_2 \in \mathcal{P}(\mathcal{U})$, we write $F_1 \ll F_2$ if F_1 is absolutely continuous with respect to F_2, and $F_1 \otimes F_2$ as the product measure. For a finite-dimensional vector, denote by $\|\cdot\|_p$ the p-norm. Denote by $L^p(F)$ the space of functions for which $\int |f|^p dF < \infty$. For a set $\mathcal{A}$, let $\operatorname{int}(\mathcal{A})$ denote its interior. Denote by $\mathbb{R}_+$ the set of non-negative real numbers, and $\mathbb{N}$ the set of natural numbers. Finally, let $\mathcal{J} := \{1, \cdots, J\}$.

2 Methodology for Time-Homogeneous Models

This section presents our methodology for time-homogeneous models. Section 2.1 defines the perturbation set. Section 2.2 gives two examples. Section 2.3 presents the general framework and derives duality results. Section 2.4 discusses how to perturb the stationary distribution.

2.1 Definition of Perturbation Set

In time-homogeneous models, we perturb the reference latent variable dynamics while imposing that the perturbed process remains stationary. Consider an unobserved stationary first-order Markov process $\{\xi_t\}_{t\in\mathbb{Z}}$ with state space $\Xi \subseteq \mathbb{R}^{d_\xi}$. The process is stationary with respect to $\nu_0 \in \mathcal{P}(\Xi)$ if and only if:

$$\int_\Xi F_0(A \mid \xi)\, \nu_0(d\xi) = \nu_0(A) \qquad \text{for all measurable } A \subseteq \Xi$$

where $F_0(d\xi'|\xi)$ is the reference transition kernel (e.g., conditional Gaussian distribution for an AR(1) process). This is equivalent to requiring that the marginal distributions of the joint distribution $dF_0(\xi, \xi') := F_0(d\xi'|\xi)\nu_0(d\xi)$ are both ν_0, i.e., $F_0 \in \Pi(\nu_0, \nu_0)$ where $\Pi(\nu_0, \nu_0)$ is the set of joint distributions on Ξ^2 with marginals ν_0. Moreover, for any $F \in \Pi(\nu_0, \nu_0)$, its conditional density $F(d\xi'|\xi)$ preserves ν_0 as a stationary distribution. Then, the perturbation set around the reference joint distribution F_0 is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\Xi^2) \mid F \in \underbrace{\Pi(\nu_0, \nu_0)}_{\text{Stationarity}}, \underbrace{D_{KL}(F\|F_0) \leq \delta}_{\text{Perturbation}}\}$$

where $\delta \geq 0$ measures the size of the Kullback-Leibler (KL) divergence ball around F_0, and

$$D_{KL}(F\|F_0) := \begin{cases} \int \log\left(\frac{dF}{dF_0}\right) dF & \text{if } F \ll F_0 \\ +\infty & \text{otherwise} \end{cases}$$

This construction imposes stationarity through a marginal constraint on the joint law, rather than through an invariant-measure restriction on an unknown transition kernel. Combined with the KL constraint, these marginal restrictions form a computationally tractable EOT problem whose implementation relies on its dual formulation (see Section 2.3).[1]

The resulting perturbation set allows us to perturb the transition kernel of the Markov process while preserving the stationary distribution by construction. It also permits nonlinear dynamics: for example, if the reference model is an AR(1) process, the perturbation set includes nonlinear first-order Markov processes with the same stationary distribution as the AR(1) process. Section 2.4 discusses how to perturb the stationary distribution. In that case, we replace $F \in \Pi(\nu_0, \nu_0)$ with $F \in \Pi(\nu, \nu)$, where ν is a perturbed stationary distribution in a neighborhood of ν_0.

The preceding construction is the $k = 2$ case of the following general formulation. We partition a vector of latent variables $U \in \mathcal{U} \subseteq \mathbb{R}^d$ into $2 \leq k \leq d$ subvectors, i.e., $U = (U_1, \cdots, U_k)$.[2] Each subvector $U_i \in \mathcal{U}_i \subseteq \mathbb{R}^{d_i}$ has a marginal distribution $\nu_i \in \mathcal{P}(\mathcal{U}_i)$ for $i = 1, \cdots, k$. Let F_0 denote the reference distribution for U. The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \cdots, \nu_k), D_{KL}(F\|F_0) \leq \delta\}$$

where $\Pi(\nu_1, \cdots, \nu_k)$ is the set of joint distributions on $\mathcal{U}$ with marginals $\{\nu_i\}_{i=1}^k$.

Remark 1. (i) In the time-homogeneous example, we take $U = (\xi, \xi')$. This pair of

[1] For duality results for general divergence-constrained OT problems, see Bayraktar et al. (2025).

[2] The vector U can also contain observable variables.

latent variables can be further partitioned into some subvectors to analyze sensitivity to assumptions about cross-sectional dependence.

(ii) Higher-order Markov dynamics can be handled by expanding the state space. For example, if the perturbed process is second-order Markov, then the perturbation set can be defined on the joint distribution of $(\tilde{\xi}_t, \tilde{\xi}_{t+1})$ where $\tilde{\xi}_t = (\xi_t, \xi_{t+1})$.

2.2 Examples

Our framework applies to a variety of latent variables, including utility shocks, productivity characteristics, labor supply shocks, etc. This section focuses on a parametric model for latent variables beyond utility shocks in infinite and finite-horizon DDC models. Appendix A considers: (i) serial independence of utility shocks in DDC models, and (ii) serial independence of consumption shocks in dynamic discrete-continuous choice models. The bounds on a scalar parameter are solutions to constrained optimization problems over the perturbation set subject to structural constraints (e.g., Bellman equation) and moment conditions.

Example 1 (Infinite Horizon Dynamic Discrete Choice Models with Serially Correlated Latent Variables)**.** This example considers serially correlated latent variables in a single-agent DDC model as in Rust (1994). Let $\xi \in \Xi$ be the exogenously evolving latent variable (e.g., unobserved productivity characteristics). Serial dependence captures the persistence in these unobserved characteristics. An AR(1) process is often used to model the transition of ξ. Let $U := (\xi, \xi')$, and let the reference distribution be $dF_0(U) := F_0(d\xi'|\xi)\nu_0(d\xi)$ where $F_0(d\xi'|\xi)$ is the reference transition kernel (e.g., the estimated AR(1) transition), and ν_0 is its stationary distribution. The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

Agents solve the smoothed[3] Bellman equation for the conditional value function $v \in \mathcal{V}$ where $\mathcal{V}$ is a function class (e.g., the class of square integrable functions): for $\forall\ (j, x, \xi) \in \mathcal{J} \times \mathcal{X} \times \Xi$, it holds that

$$v_j(x, \xi) = u_j(x, \xi; \theta) + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x,j} \left[\log \left(\sum_{j' \in \mathcal{J}} \exp(v_{j'}(x', \xi')) \right) \right] + \beta\gamma \tag{1}$$

[3]We assume that the utility shock is additively separable in the period utility function and follows an i.i.d. Extreme Value Type I distribution, leading to the log-sum-exp form of the value function Rust (1987).

where $x \in \mathcal{X}$ is the observable state variable, $\beta \in (0,1)$ is the discount factor, γ is the Euler constant, and $u_j(x, \xi; \theta)$ is the period utility of choosing $j \in \mathcal{J}$ parameterized by $\theta \in \Theta$.

Suppose the scalar parameter of interest is the average elasticity of action j with respect to variable x_l, defined as:

$$\mathbb{E}_{\nu_0}\mathbb{E}_x\left[\frac{\partial p(j|x,\xi)}{\partial x_l}\frac{x_l}{P_0(j|x)}\right]$$

where $P_0(j|x)$ is the population CCP, the expectation is taken with respect to ν_0 for ξ and the stationary distribution of x, and $p(j|x,\xi) = \frac{\exp(v_j(x,\xi))}{\sum_{j' \in \mathcal{J}} \exp\left(v_{j'}(x,\xi)\right)}$ is the model-implied Conditional Choice Probability (CCP).

We convert the smoothed Bellman equation (1) into an unconditional moment restriction that depends on the joint distribution F. We assume[4] there exists a class of test functions $\mathcal{G}$[5], chosen independently of F, such that for any $F \in \mathcal{F}$, v solves the Bellman equation (1) if and only if:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,j,x'}\left[g_j(x,\xi)\left(v_j(x,\xi) - u_j(x,\xi;\theta) - \beta \log\left(\sum_{j' \in \mathcal{J}} \exp(v_{j'}(x',\xi'))\right) - \beta\gamma\right)\right] = 0$$

where $(\xi, \xi') \sim F$, and the inner expectation is taken over the observable variables: x drawn from its stationary distribution, $j \sim P_0(j|x)$, and x' from the transition distribution conditional on (x, j). The class $\mathcal{G}$ is assumed to be rich enough under every admissible F so that the unconditional restriction is equivalent to the corresponding conditional structural equation. Let $g := (g_j)_{j \in \mathcal{J}}$. Then, we rewrite the structural constraints as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F\left[\psi(U; \theta, v, g)\right] = 0$$

We consider the following moment conditions for estimation:

$$\mathbb{E}_{\nu_0}\left[p(j|x,\xi)\right] = P_0(j|x) \quad \forall\ (j,x) \in \mathcal{J} \times \mathcal{X}$$

We assume $\mathcal{X}$ has discrete support, and rewrite the moment conditions as $\mathbb{E}_F\left[m(U; v)\right] = P_0$ where $m(U; v)$ stacks the model-implied CCPs, and P_0 stacks the population CCPs.

[4]The test function converts the continuum of conditional moment restrictions into a single unconditional moment restriction (see for example Andrews and Shi (2013) and Schennach (2014)).

[5]For example, if $\mathcal{V}$ is the class of square integrable functions, $\mathcal{G}$ is the class of square integrable functions.

Then, the lower bound on the elasticity is given by:

$$\begin{aligned}\inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} \quad & \mathbb{E}_{\nu_0}\mathbb{E}_x\left[\frac{\partial p(j|x,\xi)}{\partial x_l}\frac{x_l}{P_0(j|x)}\right] \\ \text{s.t.} \quad & \mathbb{E}_F\left[m(U;v)\right] = P_0 \\ & \sup_{g\in\mathcal{G}}\mathbb{E}_F\left[\psi(U;\theta,v,g)\right] = 0\end{aligned}$$

In the next section, we discuss the implementation for this fixed reference distribution using the algorithm proposed in Section 6.2.

Example 2 (Finite Horizon Dynamic Discrete Choice Models with Serially Correlated Latent Variables)**.** This example considers a finite-horizon DDC model where the latent variable $\xi \in \Xi$ (e.g., health shocks) follows a first-order stationary Markov process. Let $U := (\xi, \xi')$ be a vector of current and future latent variables. In practice, we may set the reference distribution as the estimated distribution from a parametric model, such as an AR(1) process. The reference distribution F_0 is the product of the conditional distribution and its stationary distribution, ν_0. The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F\|F_0) \leq \delta\}$$

The conditional value function $v_t \in \mathcal{V}$ at time period $t \leq T < +\infty$ solves the smoothed Bellman equation: for $\forall\ (x_t, \xi_t, j) \in \mathcal{X} \times \Xi \times \mathcal{J}$,

$$v_{jt}(x_t,\xi_t) = u_j(x_t,\xi_t;\theta) + \beta\mathbb{E}_{\xi_{t+1}|\xi_t}\mathbb{E}_{x_{t+1}|x_t,j}\left[\log\left(\sum_{j'\in\mathcal{J}}\exp(v_{j't+1}(x_{t+1},\xi_{t+1}))\right)\right] + \beta\gamma \qquad (2)$$

where $x \in \mathcal{X}$ is the observable state, $\beta \in (0,1)$ is the discount factor, γ is the Euler constant, $u_j(x,\xi;\theta)$ is the period utility parameterized by $\theta \in \Theta$, and $v_{jT}(x_T,\xi_T) = u_j(x_T,\xi_T;\theta)$.

Let $V_t(x_t,\xi_t) := \log\left(\sum_{j\in\mathcal{J}}\exp(v_{jt}(x_t,\xi_t))\right)$ denote the ex-ante value function at t. Suppose the scalar parameter of interest is the consumer surplus derived from the choice set $\mathcal{J}$ at period t:

$$\mathbb{E}_{\nu_0}\mathbb{E}_{x_t}\left[\frac{1}{\alpha}V_t(x_t,\xi_t)\right]$$

where we assume $u_j(x_t,\xi_t;\theta)$ is linear in price and α is the price coefficient.

We convert the smoothed Bellman equation (2) into restrictions that depend on the joint distribution $F \in \mathcal{F}$. We assume there exists a class of test functions $\mathcal{G}$, chosen independently of F, such that for any $F \in \mathcal{F}$ and each $t \leq T-1$, v_t solves (2) if and only if:

$$\sup_{g_t \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x_t, j_t, x_{t+1}} \Big[g_{j_t t}(x_t, \xi_t) \times \Big(v_{j_t t}(x_t, \xi_t) - u_{j_t}(x_t, \xi_t; \theta) - \beta V_{t+1}(x_{t+1}, \xi_{t+1}) - \beta\gamma \Big) \Big] = 0$$

The class $\mathcal{G}$ is assumed to be rich enough under every admissible F so that the unconditional restriction is equivalent to the corresponding conditional structural equation. where $(\xi_t, \xi_{t+1}) \sim F$, and the inner expectation is taken over the observable variables: x_t drawn from its period-t marginal distribution, $j_t \sim P_{0t}(j|x_t)$ from the population CCPs, and x_{t+1} from the transition distribution conditional on (x_t, j_t). Let $g := (g_{jt})_{j \in \mathcal{J}, t \leq T-1}$ and $v := (v_{jt})_{j \in \mathcal{J}, t \leq T-1}$. Then, we rewrite the structural constraints as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U; \theta, v, g) \right] = 0$$

where ψ is the sum of the objective functions in the above equation for each $t \leq T-1$.

We consider the following moment conditions for estimation: for each $t \leq T$,

$$\mathbb{E}_{\nu_0} \left[p_t(j|x_t, \xi) \right] = P_{0t}(j|x_t) \quad \forall\ (j, x_t) \in \mathcal{J} \times \mathcal{X}$$

where $p_t(j|x, \xi) = \frac{\exp(v_{jt}(x,\xi))}{\sum_{j' \in \mathcal{J}} \exp\left(v_{j't}(x,\xi)\right)}$ is the model-implied CCP at period t. We assume $\mathcal{X}$ has discrete support, and rewrite the moment conditions as $\mathbb{E}_F \left[m(U; v) \right] = P_0$ where $m(U; v)$ and P_0 stack the model-implied and population CCPs for all $t \leq T$.

Then, the lower bound on consumer surplus at period t is given by:

$$\begin{aligned} \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}} \ & \mathbb{E}_{\nu_0} \mathbb{E}_{x_t} \left[\frac{1}{\alpha} V_t(x_t, \xi_t) \right] \\ \text{s.t.} \quad & \mathbb{E}_F \left[m(U; v) \right] = P_0 \\ & \sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U; \theta, v, g) \right] = 0 \end{aligned}$$

2.3 Framework and Duality

We now present a general framework that nests the above examples. In general, the model is not point-identified when $\mathcal{F}$ is not a singleton.[6] Therefore, we propose to compute upper and lower bounds on the outcome of interest. Let the scalar parameter of interest, $s : \mathcal{U} \times \Theta \times \mathcal{V} \to \mathbb{R}$, be a function of the latent variable U, and the model primitives $(\theta, v) \in \Theta \times \mathcal{V}$. The lower

[6]For example, see Schennach (2014); Molinari (2020).

bound is the solution to the following optimization problem:

$$\begin{aligned}\kappa(\delta, P) := &\inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} \mathbb{E}_F\left[s(U;\theta,v)\right] \\ &\text{s.t.} \quad \mathbb{E}_F\left[m(U;\theta,v)\right] = P \\ &\qquad\quad \sup_{g\in\mathcal{G}} \mathbb{E}_F\left[\psi(U;\theta,v,g)\right] = 0\end{aligned} \tag{Primal}$$

where $\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \cdots, \nu_k), D_{KL}(F\|F_0) \leq \delta\}$.

The first constraint is a moment condition where the moment function $m : \mathcal{U}\times\Theta\times\mathcal{V} \to \mathbb{R}^{d_P}$ is finite-dimensional as we assume the observable variable $X \in \mathcal{X}$ has discrete support and stack the moment functions for each $x \in \mathcal{X}$. The second constraint is a structural constraint defined by $\psi : \mathcal{U}\times\Theta\times\mathcal{V}\times\mathcal{G} \to \mathbb{R}$ that is linear in the test function $g \in \mathcal{G}$ for a given (θ, v). Finally, $v \in \mathcal{V}$ is the solution to the structural constraint.

Remark 2. (i) The upper bound can be obtained by replacing $s(U;\theta,v)$ with $-s(U;\theta,v)$.

(ii) The moment condition can contain restrictions linear in F, e.g., covariance restrictions.

(iii) If some model primitives (e.g., a subvector of θ) are point-identified, they are treated as fixed values rather than optimized over.

(iv) The framework can also be applied to models without structural constraints, such as static and panel discrete choice models.

The Primal problem can be intractable due to the optimization over $\mathcal{F}$. First, the properties (e.g., closed-form and smoothness) of the optimal F^* are typically unknown, complicating the choice of approximation methods. Second, the marginal distribution conditions are functional constraints imposed directly on the distribution being optimized, which are computationally difficult to impose. Third, expectations are taken with respect to the perturbed distribution, making numerical integration difficult.

To overcome these issues, we derive the Dual problem corresponding to the Primal. The Dual provides the closed-form of the optimal F^*, and characterizes its smoothness. Moreover, in the dual, the expectation is taken with respect to the reference distribution F_0. Section 6.2 proposes a computationally tractable algorithm that utilizes the optimal F^*. To motivate the duality, consider the Lagrangian of the Primal:

$$\kappa(\delta, P) = \inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} \sup_{\lambda\in\mathbb{R}^{d_P}, g\in\mathcal{G}} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] - \lambda^T P \tag{3}$$

where $c(U;\theta,v,g,\lambda) := s(U;\theta,v) + \lambda^T m(U;\theta,v) + \psi(U;\theta,v,g)$, λ is the Lagrange multiplier for the moment condition, and g is the Lagrange multiplier function for the structural constraint under Assumption 1(iii). For a given (θ, v), under Assumption 1, we can swap the order of the infimum over F and the supremum over (λ, g). Then, we can rewrite (3) as:

$$\inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},g\in\mathcal{G}} \inf_{F\in\mathcal{F}} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] - \lambda^T P$$

whose Lagrangian is given by:

$$\inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},g\in\mathcal{G}} \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \sup_{\lambda_{KL}\geq 0} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] + \lambda_{KL}(D_{KL}(F\|F_0) - \delta) - \lambda^T P$$

where $\lambda_{KL} \geq 0$ is the Lagrange multiplier for the KL constraint. Since Slater's condition holds by Assumption 1(ii), scalar convex Lagrangian duality gives no duality gap and we can swap the order of the infimum over F and the supremum over λ_{KL}:

$$\inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},g\in\mathcal{G},\lambda_{KL}\geq 0} \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] + \lambda_{KL}(D_{KL}(F\|F_0) - \delta) - \lambda^T P$$

The inner infimum is known as the Entropic Optimal Transport (EOT) problem (for $\lambda_{KL} > 0$)[7]:

$$\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) := \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] + \lambda_{KL} D_{KL}(F\|F_0)$$

where $c(U;\theta,v,g,\lambda)$ is the cost function, and $\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL})$ is called the optimal EOT value. The EOT problem is computationally fast to solve using the Sinkhorn algorithm, which relies on the duality of the EOT problem. Section 6.1 reviews the EOT duality and the Sinkhorn algorithm. Moreover, the closed-form and smoothness of the unique solution F^* to the EOT problem can be derived from its duality (see Theorem 1 and Proposition 1). Next, we impose assumptions for the minimax theorem to swap the order of infimum and supremum, and the EOT duality to hold:

Assumption 1. *We assume:*

(i) *The marginals* $\{\nu_i\}_{i=1}^k$ *have finite p-th moment for some integer* $p \geq 1$.

(ii) $F_0 \in \Pi(\nu_1,\ldots,\nu_k)$, $F_0 \ll F_\otimes := \otimes_{i=1}^k \nu_i$, $F_\otimes \ll F_0$, *and let* $\rho(U) := \log \frac{dF_\otimes(U)}{dF_0(U)}$.

[7]See Nutz (2021) for a comprehensive introduction to the EOT problem. It is called the Optimal Transport (OT) problem when $\lambda_{KL} = 0$ (see Villani et al., 2009). The optimal value is called the optimal OT value.

(iii) *$\mathcal{G}$ is convex and symmetric, i.e., for $g_1, g_2 \in \mathcal{G}$, $\eta g_1 + (1-\eta) g_2 \in \mathcal{G}$ for $\forall\ \eta \in [0,1]$, and $-g \in \mathcal{G}$ if $g \in \mathcal{G}$. Moreover, if $g \in \mathcal{G}$, then $\eta g \in \mathcal{G}$ for $\forall\ \eta \geq 0$.*

(iv) *For $\forall\ (\theta, v, g) \in \Theta \times \mathcal{V} \times \mathcal{G}$, it holds that $\rho(U), s(U;\theta,v), \psi(U;\theta,v,g)$ are lower semi-continuous in U. Moreover, $m(U;\theta,v)$ is continuous in U.*

(v) *For $\forall\ (\theta, v, g) \in \Theta \times \mathcal{V} \times \mathcal{G}$, there exist a finite positive constant $C_{\theta,v,g}$ and $\hat{U} \in \mathcal{U}$ such that for $\forall\ U \in \mathcal{U}$, it holds that $|\rho(U)| + |s(U;\theta,v)| + \|m(U;\theta,v)\|_1 + |\psi(U;\theta,v,g)| \leq C_{\theta,v,g}(1 + d(U,\hat{U}))$ where $d(U,\hat{U}) := \sum_{i=1}^k d_i(U_i,\hat{U}_i)^p$ and d_i is a metric on $\mathcal{U}_i$.*

Assumptions 1(i)-1(iv) are mild. There is no particular necessity to write $\rho(U)$ in log-density form in Assumption 1(ii). Our notation is chosen to simplify the expression in Assumptions 1(iv) and 1(v). Assumption 1(iv) requires m to be continuous in U to ensure that $\lambda^T m(U;\theta,v)$ is lower semicontinuous in U as λ is unconstrained. Assumption 1(v) imposes the growth rate condition. It is satisfied for all Examples in Section 2.2 if we assume u, g, and v satisfy the growth rate condition. It ensures that $c(U;\theta,v,g,\lambda) \in L^1(F)$ for $\forall\ F \in \mathcal{F}$, and can also be used to show the convergence of the Sinkhorn algorithm and the convergence of optimal EOT value to optimal OT value as $\lambda_{KL} \downarrow 0$ (Eckstein and Nutz, 2022, 2024).[8]

Theorem 1. *Let $c(U;\theta,v,g,\lambda) := s(U;\theta,v) + \lambda^T m(U;\theta,v) + \psi(U;\theta,v,g)$ where $\lambda \in \mathbb{R}^{d_P}$. Under Assumption 1, for every $\delta > 0$, the following holds:*

(i) *(Minimax Duality)*

$$\kappa(\delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},\lambda_{KL}\geq 0, g\in\mathcal{G}} \mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) - \lambda_{KL}\delta - \lambda^T P \qquad \text{(Dual)}$$

where $\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL})$ is the EOT problem with regularization parameter λ_{KL}:

$$\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) := \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] + \lambda_{KL} D_{KL}(F\|F_0)$$

(ii) *(Entropic Optimal Transport Duality) For $\lambda_{KL} > 0$, we have:*[9]

$$\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) = \sup_{\{\phi_i\in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i}\phi_i(U_i) - \lambda_{KL}\mathbb{E}_{F_0}\exp\left(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}}\right) + \lambda_{KL}$$

[8]For the convergence of optimal EOT value to optimal OT value, lower semicontinuity alone is not sufficient (Nutz, 2021, Example 5.1). A sufficient condition is the continuity condition on the cost function.

[9]See Villani (2021) for optimal transport duality ($\lambda_{KL} = 0$).

Moreover, there are unique maximizers $\{\phi_i^*\}_{i=1}^k$ *up to additive constants* F_0*-almost surely, and the unique worst-case distribution* F^* *has the density of the form:*

$$\frac{dF^*(U)}{dF_0(U)} = \exp(\frac{\sum_{i=1}^k \phi_i^*(U_i) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}}) \quad F_0\text{-}a.s.$$

Furthermore, we have $\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) = \sum_{i=1}^k \mathbb{E}_{\nu_i}\phi_i^*(U_i)$*.*

(iii) If the lower semicontinuity in Assumption 1(iv) is strengthened to continuity, then optimizing over $\lambda_{KL} > 0$ *is equivalent to optimizing over* $\lambda_{KL} \geq 0$*.*

Theorem 1 establishes the duality and provides the closed-form of F^*. ϕ_i is a test function for the marginal distribution condition ν_i for $i = 1, \cdots, k$. The optimal $\{\phi_i^*\}_{i=1}^k$ are the optimal EOT potentials, which can be efficiently obtained using the Sinkhorn algorithm.

The Dual is computationally tractable. First, it provides the closed-form of the worst-case distribution F^*, which can be used in the Primal problem even if we want to solve it directly. In Section 6.2, we propose an iterative algorithm that alternates between solving the EOT to obtain the worst-case distribution and updating v by solving the structural constraint with the worst-case distribution. Second, for a given $(\theta,v,g,\lambda,\lambda_{KL})$, the optimal $\{\phi_i^*\}_{i=1}^k$ (and thus F^*) can be efficiently computed using the Sinkhorn algorithm, which is computationally very fast. Third, the expectations in the Dual are taken with respect to the marginal distributions and the reference distribution. Therefore, numerical integration methods can be determined in advance. Finally, we have:

Proposition 1 (Smoothness of the EOT potentials)**.** *Suppose* $\lambda_{KL} > 0$*. Let*

$$\tilde{c}(U;\theta,v,g,\lambda,\lambda_{KL}) := c(U;\theta,v,g,\lambda) + \lambda_{KL}\rho(U).$$

Assume $c(U;\theta,v,g,\lambda)$ *and* $\rho(U)$ *are* q*-times continuously differentiable in* U*. For each* $j \leq k$*, define the Schrödinger integrand*

$$H_j(u_j,u_{-j}) := \exp\left(\frac{\sum_{i\neq j}\phi_i^*(u_i) - \tilde{c}(u_j,u_{-j};\theta,v,g,\lambda,\lambda_{KL})}{\lambda_{KL}}\right).$$

Assume that, for each $j \leq k$*, each compact set* $K \subset \mathcal{U}_j$*, and each multi-index* a *with* $|a| \leq q$*, there exists an* $F_{\otimes,-j}$*-integrable function* $M_{j,K,a}(u_{-j})$ *such that*

$$\sup_{u_j\in K}\left|\partial_{u_j}^a H_j(u_j,u_{-j})\right| \leq M_{j,K,a}(u_{-j}) \quad F_{\otimes,-j}\text{-}a.s., \qquad \int M_{j,K,a}(u_{-j})dF_{\otimes,-j}(u_{-j}) < +\infty.$$

Then the EOT potentials admit versions $\{\phi_i^*\}_{i=1}^k$ such that ϕ_i^* is q-times continuously differentiable in U_i for each i. Consequently, $\frac{dF^*(U)}{dF_0(U)}$ admits a q-times continuously differentiable version in U.

Proposition 1 shows the smoothness of $\{\phi_i^*\}_{i=1}^k$ for $\lambda_{KL} > 0$ (the EOT case). For $\lambda_{KL} = 0$ (the OT case), it is not straightforward to obtain the smoothness of the worst-case distribution (see Villani et al., 2009, Chapter 12).

2.4 Perturbation of Stationary Distribution

This section discusses how to perturb the stationary distribution in the time-homogeneous setting. We consider the serial dependence of the latent variables, as detailed in the example in Section 2.1. Let $U := (\xi, \xi')$ be the vector of current and future latent variables. We define the perturbation set for the stationary distribution as $\mathcal{N} = \{\nu \in \mathcal{P}(\Xi) \mid D_{KL}(\nu \| \nu_0) \leq \delta_1\}$ where $\delta_1 \geq 0$. The perturbation set for the joint distribution is:

$$\mathcal{F} := \left\{F \in \mathcal{P}(\Xi^2) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}, D_{KL}(F \| F_0) \leq \delta\right\}$$

Let $\kappa_{stationary}(\delta_1, \delta, P)$ denote the lower bound on the scalar parameter of interest under this perturbation set. Under the regularity conditions stated in Theorem 2, we have:

$$\kappa_{stationary}(\delta_1, \delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P}, \lambda_{KL}\geq 0, g\in\mathcal{G}} \inf_{\nu\in\mathcal{N}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) - \lambda_{KL}\delta - \lambda^T P$$

where $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu)$ is the EOT problem with respect to the perturbed stationary distribution ν: $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) := \inf_{F\in\Pi(\nu,\nu)} \mathbb{E}_F\left[c(\xi, \xi'; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0)$. Its dual formulation is:

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) = \sup_{\phi_1,\phi_2\in L^\infty(\Xi)} \mathbb{E}_\nu\left[\phi_1(\xi) + \phi_2(\xi)\right] - \lambda_{KL}\mathbb{E}_{F_0} \exp\left(\frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}}\right) + \lambda_{KL}$$

Under regularity conditions, we can swap the order of infimum over ν and the supremum over (ϕ_1, ϕ_2). Then, the inner infimum over ν is:

$$\inf_{\nu\in\mathcal{N}} \mathbb{E}_\nu\left[\phi_1(\xi) + \phi_2(\xi)\right]$$

which is a KL-divergence distributionally robust optimization problem (see Hu and Hong, 2013; Rahimian and Mehrotra, 2019) whose dual formulation is:

$$\sup_{\eta \geq 0} -\eta \log \mathbb{E}_{\nu_0} \exp\left(-\frac{\phi_1(\xi) + \phi_2(\xi)}{\eta}\right) - \eta\delta_1$$

where η is the Lagrange multiplier for the KL divergence constraint on ν. To summarize:

Theorem 2. *Suppose Ξ is compact, $\delta > 0$, and $F_0 \in \Pi(\nu_0, \nu_0)$. For each $\nu \in \mathcal{N}$, let $F_{\otimes,\nu} := \nu \otimes \nu$ and $\rho_\nu(\xi, \xi') := \log \frac{dF_{\otimes,\nu}}{dF_0}(\xi, \xi')$. Suppose Assumptions 1(i) and 1(iii) to 1(v) hold pointwise for each $\nu \in \mathcal{N}$, with $F_{\otimes,\nu}$ and ρ_ν replacing $F_\otimes$ and ρ. Assume further that $F_{\otimes,\nu} \ll F_0$ and $D_{KL}(F_{\otimes,\nu} \| F_0) < +\infty$ for each $\nu \in \mathcal{N}$. Then, we have:*

$$\begin{aligned}\kappa_{stationary}(\delta_1, \delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\substack{\lambda\in\mathbb{R}^{d_P},\eta\geq 0\\ \lambda_{KL}\geq 0, g\in\mathcal{G}\\ \phi_1,\phi_2\in L^\infty(\Xi)\\ \phi_1,\phi_2\ l.s.c.}} & -\eta \log \mathbb{E}_{\nu_0} \exp\left(-\frac{\phi_1(\xi) + \phi_2(\xi)}{\eta}\right) - \eta\delta_1 + \lambda_{KL} - \lambda_{KL}\delta - \lambda^T P \\ & - \lambda_{KL}\mathbb{E}_{F_0} \exp\left(\frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}}\right)\end{aligned}$$

Theorem 2 shows the dual formulation when the stationary distribution is perturbed. Although the Sinkhorn algorithm does not apply directly, we can sequentially update (ϕ_1, ϕ_2) using the first-order optimality conditions like the Sinkhorn algorithm (see Section 6.1 for a review of the Sinkhorn algorithm).

3 Methodology for Time-Inhomogeneous Models

This section extends our framework to the time-inhomogeneous setting. We begin by defining the perturbation set for these models in Section 3.1. Section 3.2 provides an example of a finite-horizon dynamic discrete choice model. Section 3.3 presents the duality result. Finally, Section 3.4 discusses how to perturb the initial distribution.

3.1 Definition of Perturbation Set

Consider a sequence of latent variables over a finite horizon, $U := (\xi_1, \xi_2, \cdots, \xi_T)$, that follows a first-order Markov chain. The reference joint distribution is the product of an initial distribution and transition kernels:

$$dF_0(U) = \nu_1(d\xi_1)F_1(d\xi_2|\xi_1) \cdots F_{T-1}(d\xi_T|\xi_{T-1})$$

where $\nu_1(d\xi_1)$ is the initial distribution, and $F_t(d\xi_{t+1}|\xi_t)$ is the transition kernel from period t to $t+1$. Let $\nu_T(d\xi_T)$ be the terminal distribution implied by this process.

We consider perturbing the reference distribution while holding its initial and terminal distributions fixed, i.e.,

$$\mathcal{F}_{\text{Markov}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi_{\text{Markov}}(\nu_1, \nu_T), D_{KL}(F \| F_0) \leq \delta\}$$

where $\Pi_{\text{Markov}}(\nu_1, \nu_T)$ is the set of all joint distributions over $\mathcal{U}$ that satisfy the first-order Markov property[10] and have ν_1 and ν_T as their initial and terminal marginal distributions. Our perturbation set allows for any transition dynamics of the latent variables between the initial and terminal periods, as long as the overall process remains Markovian and the initial and terminal distributions are fixed.

We will discuss how to perturb the initial distribution in Section 3.4. The terminal distribution can often be nonparametrically identified; therefore, we fix it. For instance, in a finite horizon DDC model, the terminal period is a static discrete choice problem where the distribution of the latent variable can be identified (Lewbel, 2000; Matzkin, 2007). Moreover, if ξ_t is the market-level latent variable, and the model has the finite dependence property (see Arcidiacono and Miller, 2011)[11], then the utility parameters can be identified without a distributional assumption for the latent variables (see Kalouptsidi et al., 2021c), which in turn identifies the terminal distribution.

3.2 Example

Example 3 (Finite Horizon DDC with Time-Inhomogeneous Transition of Latent Variables)**.** This example considers a time-inhomogeneous transition for the latent variables, $U := (\xi_1, \xi_2, \cdots, \xi_T)$, whose perturbation set is $\mathcal{F}_{\text{Markov}}$. The model is similar to Example 2 but with a time-inhomogeneous transition.

We convert the smoothed Bellman equation (2) into restrictions that depend on the sequence of two-period marginal distributions $\{F_{t,t+1}\}_{t=1}^{T-1}$, where:

$$dF_{t,t+1}(\xi_t, \xi_{t+1}) = \int_{\xi_1, \ldots, \xi_{t-1}, \xi_{t+2}, \ldots, \xi_T} dF(\xi_1, \ldots, \xi_T)$$

We assume there exists a class of test functions $\mathcal{G}$, chosen independently of F, such that for

[10] That is, for any $t \leq T-1$, $F(d\xi_{t+1}|\xi_1, \cdots, \xi_t) = F(d\xi_{t+1}|\xi_t)$ almost surely under F.

[11] For example, a model with a terminating action has the finite dependence property.

any $F \in \mathcal{F}_{\text{Markov}}$ and each $t \leq T-1$, v_{jt} solves (2) if and only if:

$$\sup_{g_t \in \mathcal{G}} \mathbb{E}_{F_{t,t+1}} \mathbb{E}_{x_t, j_t, x_{t+1}} \Big[g_{j_t t}(x_t, \xi_t) \Big(v_{j_t t}(x_t, \xi_t) - u_{j_t}(x_t, \xi_t; \theta) - \beta \log \Big(\sum_{j' \in \mathcal{J}} \exp(v_{j' t+1}(x_{t+1}, \xi_{t+1})) \Big) - \beta\gamma \Big) \Big] = 0$$

The class $\mathcal{G}$ is assumed to be rich enough under every admissible F so that the unconditional restriction is equivalent to the corresponding conditional structural equation. where $(\xi_t, \xi_{t+1}) \sim F_{t,t+1}$, and (x_t, j_t) is distributed according to the observed data at time t, and x_{t+1} follows the conditional distribution given (x_t, j_t). Let $g := (g_{jt})_{j \in \mathcal{J}, t \leq T-1}$ and $v := (v_{jt})_{j \in \mathcal{J}, t \leq T-1}$. We then rewrite the structural constraints as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U; \theta, v, g) \right] = 0$$

where ψ is the sum over $t \leq T-1$ of terms inside the expectation of the previous equation.

Then, the lower bound on consumer surplus at period t is given by:

$$\begin{aligned} \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\text{Markov}}} \quad & \mathbb{E}_{\nu_t} \mathbb{E}_{x_t} \left[\frac{1}{\alpha} \log \left(\sum_{j \in \mathcal{J}} \exp(v_{jt}(x_t, \xi_t)) \right) \right] \\ \text{s.t.} \quad & \mathbb{E}_F \left[m(U; \theta, v) \right] = P_0 \\ & \sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U; \theta, v, g) \right] = 0 \end{aligned}$$

where ν_t is the marginal distribution of ξ_t implied by F.

3.3 Framework and Duality

The bound for the time-inhomogeneous case, $\kappa_{\text{TI}}(\delta, P)$, is defined similarly to the time-homogeneous case, but with the optimization performed over $\mathcal{F}_{\text{Markov}}$:

$$\begin{aligned} \kappa_{\text{TI}}(\delta, P) := \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\text{Markov}}} \quad & \mathbb{E}_F \left[s(U; \theta, v) \right] \\ \text{s.t.} \quad & \mathbb{E}_F \left[m(U; \theta, v) \right] = P \\ & \sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U; \theta, v, g) \right] = 0 \end{aligned} \tag{TI}$$

where "TI" stands for time-inhomogeneous. To solve TI, we follow the procedure in Theorem 1. However, the set $\mathcal{F}_{\text{Markov}}$ is not necessarily convex, which prevents the proof strategy of the minimax duality in Theorem 1 from being applied directly. Therefore, we propose to solve a relaxed problem where the Markov property condition is removed. We then show that, under certain reasonable conditions, the solution to the relaxed problem is Markovian, thereby also solving the original problem. The perturbation set for the relaxed problem is defined as:

$$\mathcal{F}_{\text{relaxed}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \nu_T), D_{KL}(F \| F_0) \leq \delta\}$$

where $\Pi(\nu_1, \nu_T)$ is the set of joint distributions with initial distribution ν_1 and terminal distribution ν_T. The relaxed problem is given by:

$$\begin{aligned}
\tilde{\kappa}_{\text{TI}}(\delta, P) := &\inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\text{relaxed}}} \mathbb{E}_F \left[s(U; \theta, v)\right] \\
\text{s.t.} \quad & \mathbb{E}_F \left[m(U; \theta, v)\right] = P \\
& \sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U; \theta, v, g)\right] = 0
\end{aligned} \tag{Relaxed}$$

Under Assumption 2, we have:

$$\tilde{\kappa}_{\text{TI}}(\delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F \left[c(U; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

where $c(U; \theta, v, g, \lambda) := s(U; \theta, v) + \lambda^T m(U; \theta, v) + \psi(U; \theta, v, g)$ and λ, λ_{KL}, and g are the Lagrange multipliers for the moment condition, the KL-divergence constraint, and the structural constraint, respectively. The inner infimum is:

$$\mathcal{C}_{\text{TI}}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F \left[c(U; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0)$$

which can be rewritten as the discrete-time dynamic Schrödinger Bridge (SB) problem (see Léonard, 2013; De Bortoli et al., 2021). Because it only restricts the initial and terminal distributions, we can decompose it into two parts: the two-period marginal distribution part (the first and last period) and the conditional distribution part (the intermediate variables conditional on the first and last period). The second part is unconstrained, thereby having a closed-form solution. The first part is the static SB (or EOT) problem whose duality is similar to Theorem 1. We impose the following assumptions for the minimax duality, decomposition, static SB duality, and the Markov property:

Assumption 2. *Let $dF_0^{1,T}(\xi_1, \xi_T) := \int_{\xi_2, \ldots, \xi_{T-1}} dF_0(\xi_1, \xi_2, \ldots, \xi_T)$ be the two-period marginal of F_0 at periods 1 and T. We assume:*

(i) $\mathcal{U}$ *is compact.*

(ii) $F_0^{1,T} \sim \nu_1 \otimes \nu_T$, *i.e.,* $F_0^{1,T}$ *and* $\nu_1 \otimes \nu_T$ *are mutually absolutely continuous. Moreover,* $\log \frac{d(\nu_1 \otimes \nu_T)}{dF_0^{1,T}} \in L^1(\nu_1 \otimes \nu_T)$.

(iii) *For* $\forall\ (\theta, v, g) \in \Theta \times \mathcal{V} \times \mathcal{G}$, *there exists a finite positive constant* $C_{\theta,v,g}$ *such that for* $\forall\ U \in \mathcal{U}$, *it holds that* $|s(U;\theta,v)| + \|m(U;\theta,v)\|_1 + |\psi(U;\theta,v,g)| \leq C_{\theta,v,g}$.

(iv) *The functionals* $s(U;\theta,v)$, $m(U;\theta,v)$, *and* $\psi(U;\theta,v,g)$ *are pairwise additive, i.e.,* $s(U;\theta,v) = \sum_{t=1}^{T-1} s_t(\xi_t,\xi_{t+1};\theta,v_t,v_{t+1})$, $m(U;\theta,v) = \sum_{t=1}^{T-1} m_t(\xi_t,\xi_{t+1};\theta,v_t,v_{t+1})$, *and* $\psi(U;\theta,v,g) = \sum_{t=1}^{T-1} \psi_t(\xi_t,\xi_{t+1};\theta,v_t,v_{t+1},g_t,g_{t+1})$ *for some functions* s_t, m_t, *and* ψ_t.

The boundedness condition in Assumption 2(iii) is stronger than Assumption 1(v), which does not guarantee that $c(U;\theta,v,g,\lambda) \in L^1(F)$ for any $F \in \mathcal{F}_{\text{relaxed}}$. Assumption 2(ii) is a sufficient condition for the SB duality to hold. Finally, Assumption 2(iv) is the key to the Markov property of the solution to the Relaxed problem, and is satisfied in Example 3. It does not hold if the moment function depends on the entire path of the latent variables.

Theorem 3. *Let* $c(U;\theta,v,g,\lambda) := s(U;\theta,v) + \lambda^T m(U;\theta,v) + \psi(U;\theta,v,g)$ *where* $\lambda \in \mathbb{R}^{d_P}$. *Under Assumptions 1(i), 1(iii), 1(iv), and 2, for every* $\delta > 0$, *the following holds:*

(i) *(Minimax Duality)*

$$\tilde{\kappa}_{TI}(\delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},\lambda_{KL}\geq 0, g\in\mathcal{G}} \mathcal{C}_{TI}(\theta,v,g,\lambda,\lambda_{KL}) - \lambda_{KL}\delta - \lambda^T P$$

where $\mathcal{C}_{TI}(\theta,v,g,\lambda,\lambda_{KL})$ *is defined as:*

$$\mathcal{C}_{TI}(\theta,v,g,\lambda,\lambda_{KL}) := \inf_{F\in\Pi(\nu_1,\nu_T)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] + \lambda_{KL} D_{KL}(F\|F_0) \qquad (\mathrm{S_{dyn}})$$

(ii) *For* $\lambda_{KL} > 0$, *the unique worst-case distribution to* S_{dyn} *has the density of the form:*

$$\frac{dF^*(U)}{dF_0(U)} = \exp\left(\frac{\phi_1^*(\xi_1) + \phi_T^*(\xi_T) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}}\right)$$

where $\phi_1^*(\xi_1)$ *and* $\phi_T^*(\xi_T)$ *are the unique maximizers (up to an additive constant) to:*

$$\sup_{\phi_1\in L^1(\nu_1),\phi_T\in L^1(\nu_T)} \mathbb{E}_{\nu_1}\phi_1(\xi_1) + \mathbb{E}_{\nu_T}\phi_T(\xi_T) - \lambda_{KL}\mathbb{E}_{R_{1,T}}\exp\left(\frac{\phi_1(\xi_1)+\phi_T(\xi_T)}{\lambda_{KL}}\right) + \lambda_{KL}$$

where the auxiliary reference measure $R_{1,T}$ *is defined as:*

$$dR_{1,T}(\xi_1,\xi_T) := \int_{\xi_2,\ldots,\xi_{T-1}} \exp\left(\frac{-c(U;\theta,v,g,\lambda)}{\lambda_{KL}}\right) dF_0(\xi_1,\ldots,\xi_T)$$

Furthermore, the solution F^* has the Markov property, i.e., $F^* \in \Pi_{Markov}(\nu_1, \nu_T)$.

(iii) (Equivalence) $\kappa_{TI}(\delta, P) = \tilde{\kappa}_{TI}(\delta, P)$.

Theorem 3 shows the duality for the Relaxed problem. The difference between Theorem 3(i) and Theorem 1(i) is that (S_{dyn}) does not fix the intermediate marginal distributions. Therefore, we can decompose (S_{dyn}) into the sum of two-period marginal distribution ($F_0^{1,T}$) part, and the conditional distribution (the distribution of $(\xi_2, \cdots, \xi_{T-1})$ given ξ_1 and ξ_T) part. The latter part is an unconstrained optimization problem, thereby having a closed-form solution. The first part is the static SB problem, whose duality is given by Theorem 3(ii).

Assumption 2(iv) is crucial for both the Markov property in Theorem 3(ii) and the equivalence between the TI and Relaxed problems. Under this assumption, the cost function $c(U; \theta, v, g, \lambda)$ is pairwise additive. Therefore, for $\lambda_{KL} > 0$, the density ratio in Theorem 3(ii) has the Markov property.[12] Moreover, Assumption 2(iv) implies that any relaxed feasible distribution can be Markovianized without changing the objective, moment restrictions, or structural restrictions, and without increasing the KL divergence. Hence the TI problem is equivalent to the Relaxed problem.

The Relaxed problem can also be solved using the iterative algorithm proposed in Section 6.2. There is an additional step to obtain the two-period auxiliary reference distribution $R_{1,T}$. The S_{dyn} can also be solved using the Sinkhorn algorithm.

3.4 Perturbation of Initial Distribution

This section extends the framework to allow sensitivity analysis with respect to the initial distribution. Let $\delta > 0$, and let $\mathcal{N}$ be a convex closed set containing the initial distribution ν_1, e.g., $\mathcal{N} = \{\nu \in \mathcal{P}(\Xi) \mid D_{KL}(\nu \| \nu_1) \leq \delta_1\}$ where $\delta_1 \geq 0$. Then, the perturbation set is

$$\mathcal{F}_{\mathcal{N},\text{Relaxed}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_T), \nu \in \mathcal{N}, D_{KL}(F \| F_0) \leq \delta\}$$

Let $\tilde{\kappa}_{\text{TI,Initial}}(\delta, P)$ be the lower bound on the scalar parameter for the Relaxed problem with the perturbation set $\mathcal{F}_{\mathcal{N},\text{Relaxed}}$. The following minimax duality similar to Theorem 3 holds:

Theorem 4 (Minimax Duality with Perturbation of Initial Distribution). *Suppose $\mathcal{N}$ is convex and closed, and that the assumptions in Theorem 3 hold for each $\nu \in \mathcal{N}$. Then,*

$$\tilde{\kappa}_{TI,Initial}(\delta, P) = \inf_{(\theta,v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_{KL} \geq 0, g \in \mathcal{G}} \mathcal{C}_{TI,Initial}(\theta, v, g, \lambda, \lambda_{KL}) - \lambda_{KL}\delta - \lambda^T P$$

[12] It can be treated as the (unnormalized) pairwise Markov random field Wainwright et al. (2008).

where $\mathcal{C}_{TI,Initial}(\theta, v, g, \lambda, \lambda_{KL})$ is defined as follows:

$$\mathcal{C}_{TI,Initial}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{\nu \in \mathcal{N}} \inf_{F \in \Pi(\nu, \nu_T)} \mathbb{E}_F \left[c(U; \theta, v, g, \lambda) \right] + \lambda_{KL} D_{KL}(F \| F_0) \tag{4}$$

To solve (4), as shown in the proof of Theorem 3(ii) and Section 3.3, we can decompose the problem into two parts: the two-period marginal distribution part, and the conditional distribution part. The first part requires solving:

$$\inf_{\nu \in \mathcal{N}} \inf_{F_{1,T} \in \tilde{\Pi}(\nu, \nu_T)} D_{KL}(F_{1,T} \| R_{1,T})$$

where $\tilde{\Pi}(\nu, \nu_T)$ is the set of distributions of (ξ_1, ξ_T) whose marginal distributions are ν and ν_T, respectively. The inner problem is the static SB problem whose duality is the same as the EOT problem (Nutz, 2021). Let $\mathrm{EOT}(\nu, \nu_T)$ be the optimal value of the inner problem.

Lemma 1. *Under the assumptions in Theorem 4, $EOT(\nu, \nu_T)$ is convex in ν. Furthermore, if $\frac{dR_{1,T}}{d(\nu \otimes \nu_T)}$ is continuous and bounded away from zero and above uniformly for $\nu \in \mathcal{N}$, then $EOT(\nu, \nu_T)$ is directionally differentiable in ν, and its directional derivative with respect to ν toward $\nu' \in \mathcal{N}$ is given by:*

$$\lim_{\epsilon \downarrow 0} \frac{EOT(\nu + \epsilon(\nu' - \nu), \nu_T) - EOT(\nu, \nu_T)}{\epsilon} = \int \phi^* d(\nu' - \nu)$$

where ϕ^* is the optimal EOT potential for ν.

Lemma 1 shows that (4) is a convex optimization problem with respect to ν. Moreover, ϕ^* is a result of the Sinkhorn algorithm, which can be used to search the optimal ν efficiently.

4 Large Sample Properties

This section establishes the large sample properties of the estimator for the bound. Section 4.1 proposes a consistent estimator and shows its convergence rate. Section 4.2 establishes the asymptotic distribution of the plug-in estimator for the bound.

4.1 Consistency and Convergence Rate

The bound on the scalar parameter is the projection of the identified set defined by the moment conditions and structural constraints onto the scalar parameter. We follow Chernozhukov et al. (2007) to propose an estimator for the identified set and show its consistency and convergence rate. Let P_n be an estimator for P_0 where n is the sample size. Denote by $\epsilon_n \in \mathbb{R}_+$ the tolerance level for the moment conditions that goes to zero at a suitable rate as $n \to \infty$. Our estimators $\kappa(\delta, P_n, \epsilon_n)$, and $\tilde{\kappa}_{\mathrm{TI}}(\delta, P_n, \epsilon_n)$ for the bounds replace the moment conditions by the approximate moment conditions.[13]

Assumption 3. *Let $\mathcal{A}$ be either $\Theta \times \mathcal{F}$ or $\Theta \times \mathcal{F}_{relaxed}$, and $\alpha := (\theta, F) \in \mathcal{A}$. Assume:*

(i) $\Theta \subseteq \mathbb{R}^{d_\theta}$ *is convex and compact.*

(ii) *If $\mathcal{A} = \Theta \times \mathcal{F}_{relaxed}$, then Assumption 2(i) holds.*

(iii) *For $\forall\ \alpha \in \mathcal{A}$, the structural constraint F-a.s. has a unique solution $v(\alpha) \in \mathcal{V}$.*

(iv) *The identified set $\mathcal{A}_I := \{\alpha \in \mathcal{A} \mid \mathbb{E}_F[m(U; \theta, v(\alpha))] = P_0\}$ is nonempty.*

(v) $\mathbb{E}_F[m(U; \theta, v(\alpha))]$ *is continuous in $\alpha \in \mathcal{A}$, i.e., the preimages of closed sets are closed.*

Assumption 3(i) is mild. Assumption 3(iii) holds in single-agent DDC models. It rules out dynamic games with multiple equilibria. Assumption 3(iv) is also mild as the identified set for θ is usually nonempty under the reference distribution F_0. Moreover, if the identified set for $\delta = +\infty$[14] is nonempty, then Assumption 3(iv) implicitly assumes the radius δ is large enough. The smallest radius such that the identified set is nonempty can be estimated (see Remark 5). Assumption 3(v) implies that the identified set and its estimator are compact as $\mathcal{F}$ is compact (see Lemma 8) and thus $\mathcal{A}$ is compact (see Lemma 2).

Under Assumption 3, the estimator for $\mathcal{A}_I$ is defined as:

$$\hat{\mathcal{A}}_I := \{\alpha \in \mathcal{A} \mid \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty \leq \epsilon_n\}$$

The analysis of the consistency and convergence rate uses the *Hausdorff Distance*:

$$d_H(\mathcal{A}_1, \mathcal{A}_2) := \max\left\{\sup_{\alpha_1 \in \mathcal{A}_1} d(\alpha_1, \mathcal{A}_2), \sup_{\alpha_2 \in \mathcal{A}_2} d(\alpha_2, \mathcal{A}_1)\right\}$$

where $d(\alpha_1, \mathcal{A}_1) := \inf_{\alpha_2 \in \mathcal{A}_2} d(\alpha_1, \alpha_2)$ and $d(\alpha_1, \alpha_2)$ is a metric on $\mathcal{A}$.

Assumption 4. *Assume P_n is a $\sqrt{n}$-consistent estimator for P_0, and there exists c_n such that $\sqrt{n}\|P_0 - P_n\|_\infty \leq c_n$ with probability approaching 1 where c_n can be data-dependent. Let $\epsilon_n = \frac{c_n}{\sqrt{n}}$, and assume $\epsilon_n \xrightarrow{p} 0$.*

[13]To compute the estimator, the number of moment conditions is doubled due to the use of approximate moment conditions. The duality is similar to Theorems 1 and 3, thus is omitted for brevity.

[14]In this case, the KL divergence constraint is replaced by the absolute continuity constraint, i.e., $F \ll F_0$.

Assumption 4 is mild as we assumed the observable variable has discrete support, e.g., P_n can be the frequency estimator. In practice, we can set $c_n \propto \log n$. Then, the convergence rate in Theorem 5(ii) is $\sqrt{n}$-consistent up to a logarithmic factor. We show some properties of the identified set and its estimator:

Lemma 2. *Under Assumptions 3 and 4, $\mathcal{A}_I, \hat{\mathcal{A}}_I$ are closed and compact. Moreover, $\hat{\mathcal{A}}_I$ is nonempty.*

By the extreme value theorem, the infimum is achieved if the scalar parameter is continuous on $\mathcal{A}$, i.e., Assumption 6(i) holds. Therefore, the optimization problem has a solution. Next, we impose the polynomial minorant condition as in Chernozhukov et al. (2007) for the convergence rate of the estimator:

Assumption 5 (Polynomial Minorant Condition)**.** *There exists positive constants C_1 and C_2 such that: $\|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_0\|_\infty \geq C_1 \min\{C_2, d(\alpha,\mathcal{A}_I)\}$.*

Theorem 5. *Under Assumptions 3 and 4, we have:*

(i) *(Consistency) $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = o_p(1)$.*

(ii) *(Convergence Rate) Under Assumption 5, $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = O_p(\frac{\max\{1,c_n\}}{\sqrt{n}})$.*

Theorem 5 establishes the $\sqrt{n}$-consistency up to a logarithmic factor (if $c_n \propto \log n$). Then, we impose the following continuity assumption on the scalar parameter of interest:

Assumption 6. *Let $s(\alpha) := \mathbb{E}_F[s(U;\theta,v(\alpha))]$, assume one of the following:*

(i) *$s(\alpha)$ is continuous in $\alpha \in \mathcal{A}$.*

(ii) *$s(\alpha)$ is Lipschitz continuous in $\alpha \in \mathcal{A}$.*

Theorem 6. *Under Assumptions 3 and 4, we have:*

(i) *(Consistency) Under Assumption 6(i), $\kappa(\delta, P_n, \epsilon_n) \xrightarrow{p} \kappa(\delta, P_0)$, and $\tilde{\kappa}_{TI}(\delta, P_n, \epsilon_n) \xrightarrow{p} \tilde{\kappa}_{TI}(\delta, P_0)$.*

(ii) *(Convergence Rate) Under Assumptions 5 and 6(ii), $|\kappa(\delta, P_n, \epsilon_n) - \kappa(\delta, P_0)| = O_p(\frac{\max\{1,c_n\}}{\sqrt{n}})$, and $|\tilde{\kappa}_{TI}(\delta, P_n, \epsilon_n) - \tilde{\kappa}_{TI}(\delta, P_0)| = O_p(\frac{\max\{1,c_n\}}{\sqrt{n}})$.*

4.2 Asymptotic Distribution

This section establishes the asymptotic distribution of $\kappa(\delta, P_n)$ and $\tilde{\kappa}_{\text{TI}}(\delta, P_n)$. To this end, we first show the Hadamard directional differentiability of $\kappa(\delta, P)$ and $\tilde{\kappa}_{\text{TI}}(\delta, P)$ with respect to P at P_0 similar to Christensen and Connault (2023). We begin with the definition of Hadamard directional differentiability:

Definition 1. *The map $f : \mathbb{R}^{d_P} \to \mathbb{R}$ is Hadamard directionally differentiable at $P \in \mathbb{R}^{d_P}$, if there exists a continuous map $f' : \mathbb{R}^{d_P} \to \mathbb{R}$ such that for $h \in \mathbb{R}^{d_P}$, we have:*

$$\lim_{i\to\infty} \frac{f(P + t_i h_i) - f(P)}{t_i} = f'(P; h)$$

for all sequences $\{h_i\} \subseteq \mathbb{R}^{d_P}$, $t_i \downarrow 0$, and $h_i \to h \in \mathbb{R}^{d_P}$ as $i \to \infty$.

Under Assumption 3, we can restate the optimization problem as:

$$\inf_{\alpha\in\mathcal{A}} s(\alpha) \quad \text{s.t.} \quad P(\alpha) = P_0$$

where $P(\alpha) := \mathbb{E}_F\left[m(U; \theta, v(\alpha))\right]$. Moreover, the identified set $\mathcal{A}_I$ is nonempty, which means the feasible set for the optimization problem is nonempty. By Lemma 2, $\mathcal{A}_I$ is compact. Under Assumption 6(i), the Extreme Value Theorem (see Rudin et al., 1976, Theorem 4.16.) implies that the infimum is attained. Denote by $\mathcal{A}^*_{I,\text{TH}}$, $\mathcal{A}^*_{I,\text{TI}}$ the nonempty sets of optimizers for the problems $\kappa(\delta, P_0)$ and $\tilde{\kappa}_{\text{TI}}(\delta, P_0)$, respectively.

To establish Hadamard directional differentiability of $\kappa(\delta, P)$ and $\tilde{\kappa}_{\text{TI}}(\delta, P)$ at P_0, we impose assumptions similar to those in Bonnans and Shapiro (2013) Theorem 4.25.[15] Since it is stated for optimization problems on Banach spaces, the perturbation analysis in this subsection is carried out in a norm topology. Let $\|\cdot\|_{\mathbb{B}}$ denote a Banach norm on the ambient space containing $\mathcal{A}$; for example, one may take $\|(\theta, F) - (\theta', F')\|_{\mathbb{B}} := \|\theta - \theta'\| + \|F - F'\|_{\text{TV}}$. Define $\text{dist}_{\mathbb{B}}(\alpha, \mathcal{A}^*_I) := \inf_{\alpha^*\in\mathcal{A}^*_I} \|\alpha - \alpha^*\|_{\mathbb{B}}$.

Assumption 7. *Assume $s(\alpha)$ and $P(\alpha)$ are continuously differentiable on $\mathcal{A}$. That is, they are Gâteaux differentiable on $\mathcal{A}$ and the corresponding derivatives $Ds(\alpha)$ and $DP(\alpha)$ are continuous on $\mathcal{A}$ (in the operator norm topology).*[16]

Assumption 8. *Let $\mathcal{A}^*_I$ be either $\mathcal{A}^*_{I,TH}$ or $\mathcal{A}^*_{I,TI}$. Assume:*

[15] We work on the primal problem to show the Hadamard directional differentiability, while Christensen and Connault (2023) works on the dual problem (see their Theorem 6.2).

[16] For a given direction $\alpha_1 \in \mathcal{A}$, the Gâteaux derivatives are understood as $Ds(\alpha)(\alpha_1 - \alpha)$ and $DP(\alpha)(\alpha_1 - \alpha)$. See Bonnans and Shapiro (2013) Page 35 for the definition of Gâteaux derivative.

(i) $0 \in int\{DP(\alpha)(\mathcal{A}-\alpha)\}$ *for* $\forall\ \alpha \in \mathcal{A}_I^*$.

(ii) *For* $\forall\ h \in \mathbb{R}^{d_P}$, *it holds that for* $\forall\ P_t := P_0 + th + o(t)$ *and* $t > 0$ *small enough, the problem* $\kappa(\delta, P_t)$ *or* $\tilde{\kappa}_{TI}(\delta, P_t)$ *has an* $o(t)$*-optimal solution* $\alpha(t)$ *such that* $\mathrm{dist}_{\mathbb{B}}(\alpha(t), \mathcal{A}_I^*) = O(t)$.

(iii) *For* $\forall\ t_n \downarrow 0$, *the sequence* $\{\alpha(t_n)\}$ *has a limit point (in the norm topology) in* $\mathcal{A}_I^*$.

Theorem 7. *Under Assumptions 3, 7, and 8, the maps* $\kappa(\delta, P)$ *and* $\tilde{\kappa}_{TI}(\delta, P)$ *are Hadamard directionally differentiable at* P_0 *in any direction* $h \in \mathbb{R}^{d_P}$, *and:*

$$\kappa'(\delta, P_0; h) = \inf_{\alpha \in \mathcal{A}_{I,TH}^*} \sup_{\lambda \in \Lambda(\alpha, P_0)} -\lambda^T h, \quad \tilde{\kappa}'_{TI}(\delta, P_0; h) = \inf_{\alpha \in \mathcal{A}_{I,TI}^*} \sup_{\lambda \in \Lambda(\alpha, P_0)} -\lambda^T h$$

where $\Lambda(\alpha, P_0)$ *is the nonempty set of Lagrange multipliers corresponding to* $\alpha \in \mathcal{A}_I^*$.[17]

Moreover, if $\sqrt{n}(P_n - P_0) \xrightarrow{d} Z \sim \mathcal{N}(0, \Sigma)$, *then* $\sqrt{n}(\kappa(\delta, P_n) - \kappa(\delta, P_0)) \xrightarrow{d} \kappa'(\delta, P_0; Z)$ *and* $\sqrt{n}(\tilde{\kappa}_{TI}(\delta, P_n) - \tilde{\kappa}_{TI}(\delta, P_0)) \xrightarrow{d} \tilde{\kappa}'_{TI}(\delta, P_0; Z)$.

Theorem 7 shows the asymptotic distribution of the bound's estimator. To conduct inference, we may follow the procedure in Fang and Santos (2019). In addition, the numerical delta method Hong and Li (2018) combined with our practical implementation in Section 6 can be used to overcome the computational challenge. See Section 7 for the details.

5 Interpreting the Results

In practice, we can estimate an alternative (parametric) model and set the radius to be the KL divergence from the alternative to the reference distribution. In addition, this section considers three complementary sensitivity measures to interpret the results: global sensitivity, local sensitivity, and robustness metric.

5.1 Global Sensitivity

The global sensitivity[18] approach removes the KL divergence constraint and considers the nonparametric bounds. We study the finite-radius entropic approximation as $\delta \to \infty$ and

[17] See Bonnans and Shapiro (2013) Definition 3.8 and Theorem 3.9. Robinson's constraint qualification is satisfied under Assumptions 3, 7, and 8 (see Appendix B.4.4). Therefore, $\Lambda(\alpha^*, P_0)$ is nonempty.

[18] See Christensen and Connault (2023) Theorem 2.1 for similar results. However, their results are silent about how large the radius should be so that the bounds are close to the nonparametric bounds.

provide an explicit pointwise upper bound on the inner approximation error. We focus on the time-homogeneous case and show that this yields a computationally tractable approximation to the nonparametric bound defined by:

$$
\begin{aligned}
\kappa_{\mathrm{NP}}(P) := \inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} & \mathbb{E}_F\left[s(U;\theta,v)\right] \\
\text{s.t.} \quad & \mathbb{E}_F\left[m(U;\theta,v)\right] = P \\
& \sup_{g\in\mathcal{G}} \mathbb{E}_F\left[\psi(U;\theta,v,g)\right] = 0
\end{aligned}
\tag{Primal}
$$

where $\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1,\cdots,\nu_k)\}$. Moreover, we provide an explicit upper bound on the approximation error.

After applying the minimax duality, we need to solve the following problem:

$$
\inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},g\in\mathcal{G}} \inf_{F\in\Pi(\nu_1,\ldots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] - \lambda^T P
$$

where the inner problem is an OT problem, which is computationally challenging in high-dimensional settings. The EOT is a computationally tractable approximation to the OT problem. Recall $\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) := \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] + \lambda_{KL} D_{KL}(F\|F_0)$.

Theorem 8 (Adapted from Eckstein and Nutz (2024) Theorem 3.1(i))**.** *Suppose Assumption 1 holds. Assume the marginals $\{\nu_i\}_{i=1}^k$ have finite $p+\eta$-th moment for some $\eta > 0$ and integer $p \geq 1$, and $c(U;\theta,v,g,\lambda)$ satisfies the $A_{L,C}$ condition in Eckstein and Nutz (2024) where L, C depend on (θ,v,g,λ). Moreover, assume $\|\log \frac{dF_\otimes}{dF_0}\|_\infty$ is finite. Let d_i be the dimension of U_i. Then, for any $\lambda_{KL} \in (0,1]$,*

$$
0 \leq \mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) - \mathcal{C}(\theta,v,g,\lambda,0) \leq \left(\sum_{i=2}^k d_i\right)\lambda_{KL}\log\left(\frac{1}{\lambda_{KL}}\right) + \left((k-1)^{\frac{1}{p}}LC + \|\log\frac{dF_\otimes}{dF_0}\|_\infty\right)\lambda_{KL}
$$

Theorem 8 provides an explicit upper bound on the approximation error of $\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL})$ to $\mathcal{C}(\theta,v,g,\lambda,0)$. For DDC models, the constants L and C can be explicitly characterized under additional conditions (see Eckstein and Nutz, 2022, Lemma 3.5), and (Eckstein and Nutz, 2024, Remark 2.1). The upper bound strictly decreases to zero as $\lambda_{KL} \downarrow 0$. Therefore, we can choose a sufficiently small λ_{KL} (or sufficiently large δ) to achieve a desired accuracy for the approximation. Our framework thus approximates the nonparametric bounds in a computationally tractable way with an explicitly quantifiable approximation error.

Remark 3. Under certain conditions, one can establish the convergence of the EOT worst-case distribution to the OT worst-case distribution as the regularization parameter $\lambda_{KL} \downarrow 0$.

Nutz (2021) Theorem 5.5 provides one sufficient condition: the existence of a solution F^* to the OT problem such that $D_{KL}(F^*\|F_0) < +\infty$.

5.2 Local Sensitivity

The local sensitivity[19] approach computes the right derivative of the bounds with respect to δ, which measures the effect of a small increase of the chosen radius on the bounds. We show the right differentiability of the bounds with respect to δ. Define:

$$\Pi_{\text{TH}} := \{F \in \Pi(\nu_1, \cdots, \nu_k) \mid C_3 \le dF \le C_4, \|dF\|_{Lip} \le L\}$$
$$\Pi_{\text{TI}} := \{F \in \Pi(\nu_1, \nu_T) \mid C_3 \le dF \le C_4, \|dF\|_{Lip} \le L\}$$

where $\|\cdot\|_{Lip}$ is the Lipschitz constant, and C_3, C_4, L are positive constants. We assume:

Assumption 9. *Let $\mathcal{A}^{\delta}_{I,Lip}$ be either $\mathcal{A}^{\delta,Lip}_{I,TH}$ or $\mathcal{A}^{\delta,Lip}_{I,TI}$ defined as:*

$$\mathcal{A}^{\delta,Lip}_{I,TH} := \{\alpha \in \Theta \times \Pi_{TH} \mid \mathbb{E}_F\left[m(U;\theta, v(\alpha))\right] = P_0, D_{KL}(F\|F_0) \le \delta\}$$
$$\mathcal{A}^{\delta,Lip}_{I,TI} := \{\alpha \in \Theta \times \Pi_{TI} \mid \mathbb{E}_F\left[m(U;\theta, v(\alpha))\right] = P_0, D_{KL}(F\|F_0) \le \delta\}$$

Let Π_{Lip} denote either Π_{TH} or Π_{TI}, corresponding to the choice of $\mathcal{A}^{\delta}_{I,Lip}$. Let $\mathcal{A}^{\delta,*}_{I,Lip}$ be either $\mathcal{A}^{\delta,Lip,*}_{I,TH}$ or $\mathcal{A}^{\delta,Lip,*}_{I,TI}$ that are the sets of solutions to the corresponding restricted optimization problems over $\mathcal{A}^{\delta,Lip}_{I,TH}$ and $\mathcal{A}^{\delta,Lip}_{I,TI}$. Assume:

(i) $\delta^* := \inf\{\delta \ge 0 \mid \mathcal{A}^{\delta}_{I,Lip} \neq \emptyset\}$ *is finite.*[20]

(ii) $\mathcal{U}$ *is compact.*

(iii) $\|dF_0\|_{Lip} \le L$, *and* $C_3 \le dF_0 \le C_4$.

(iv) $0 \in int\{DP(\alpha)(\Theta \times \Pi_{Lip} - \alpha)\}$ *for* $\forall\ \alpha \in \mathcal{A}^{\delta,*}_{I,Lip}$.

(v) *For* $\forall\ \delta_t := \delta + t + o(t)$ *and* $t > 0$ *small enough, the optimization problem corresponding to* δ_t *has an* $o(t)$*-optimal solution* $\alpha(t)$ *such that* $d(\alpha(t), \mathcal{A}^{\delta,*}_{I,Lip}) = O(t)$.

(vi) *For* $\forall\ t_n \downarrow 0$ *the sequence* $\{\alpha(t_n)\}$ *has a limit point (in the norm topology)* $\alpha_0 \in \mathcal{A}^{\delta,*}_{I,Lip}$.

[19]See Bartl et al. (2021) Theorem 2.2 and Christensen and Connault (2023) Page 276 for similar results.
[20]The smallest radius can be computed, see Remark 5.

Define the restricted values:

$$\kappa^{Lip}(\delta, P_0) := \inf_{\alpha \in \mathcal{A}^{\delta, Lip}_{I,\mathrm{TH}}} s(\alpha), \quad \tilde{\kappa}^{Lip}_{\mathrm{TI}}(\delta, P_0) := \inf_{\alpha \in \mathcal{A}^{\delta, Lip}_{I,\mathrm{TI}}} s(\alpha).$$

Theorem 9. *Under Assumptions 3, 7, and 9, $\kappa^{Lip}(\delta, P_0)$ and $\tilde{\kappa}^{Lip}_{TI}(\delta, P_0)$ are right differentiable at $\delta > \delta^*$ and their right derivatives are given by:*

$$\lim_{\epsilon \downarrow 0} \frac{\kappa^{Lip}(\delta + \epsilon, P_0) - \kappa^{Lip}(\delta, P_0)}{\epsilon} = \inf_{\alpha \in \mathcal{A}^{\delta, Lip, *}_{I,TH}} \sup_{\lambda_{KL} \in \Lambda_{KL}(\alpha, \delta)} -\lambda_{KL}$$

$$\lim_{\epsilon \downarrow 0} \frac{\tilde{\kappa}^{Lip}_{TI}(\delta + \epsilon, P_0) - \tilde{\kappa}^{Lip}_{TI}(\delta, P_0)}{\epsilon} = \inf_{\alpha \in \mathcal{A}^{\delta, Lip, *}_{I,TI}} \sup_{\lambda_{KL} \in \Lambda_{KL}(\alpha, \delta)} -\lambda_{KL}$$

where $\Lambda_{KL}(\alpha, \delta)$ is the nonempty set of Lagrange multipliers corresponding to (α, δ).

Theorem 9 shows the right differentiability. We can also compute the derivative of the length of the bounds. In practice, we may need to compute it numerically due to the optimization over the set of optimizers and the Lagrange multipliers.

Remark 4. The boundary case $\delta = 0$ is not covered by Theorem 9. At the reference model, the natural local scale for KL neighborhoods is the square-root radius. To see this, Lam (2016, Theorem 3.1) considers the following expansion for a function h of a random variable X:

$$\sup_{D_{KL}(P_f \| P_0) \leq \delta} \mathbb{E}_f[h(X)] = \mathbb{E}_0[h(X)] + \sqrt{2\,\mathrm{Var}_0(h(X))}\delta^{1/2} + O(\delta), \quad \delta \downarrow 0.$$

Thus, the ordinary right derivative with respect to the KL radius can be infinite at the reference model. This motivates writing $\delta = r^2$ and defining:

$$\bar{\kappa}^{Lip}(r, P_0) := \kappa^{Lip}(r^2, P_0), \quad \bar{\tilde{\kappa}}^{Lip}_{\mathrm{TI}}(r, P_0) := \tilde{\kappa}^{Lip}_{\mathrm{TI}}(r^2, P_0), \quad r \geq 0.$$

The square-root-scale local derivatives at the reference model are then defined, when the limits exist, by:

$$\lim_{r \downarrow 0} \frac{\bar{\kappa}^{Lip}(r, P_0) - \bar{\kappa}^{Lip}(0, P_0)}{r}, \quad \lim_{r \downarrow 0} \frac{\bar{\tilde{\kappa}}^{Lip}_{\mathrm{TI}}(r, P_0) - \bar{\tilde{\kappa}}^{Lip}_{\mathrm{TI}}(0, P_0)}{r}.$$

5.3 The Robustness Metric

The robustness metric is the smallest deviation from the reference distribution that can lead to sensitive results (Spini, 2024). In practice, we begin by estimating a reference scalar parameter, $\hat{s}_{F_0}$, under the reference distribution. If the perturbed scalar parameter s_F is below a certain threshold, e.g., $\bar{s} = 0.95 \cdot \hat{s}_{F_0}$, then we may be concerned about the robustness of the results. The robustness metric is defined as:

$$\begin{aligned}
\delta(\bar{s}, P) := \inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\Pi(\nu_1,\cdots,\nu_k)} & D_{KL}(F\|F_0) & \tilde{\delta}_{\mathrm{TI}}(\bar{s}, P) := \inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\Pi(\nu_1,\nu_T)} & D_{KL}(F\|F_0) \\
\text{s.t.} \quad & \mathbb{E}_F\left[m(U;\theta,v)\right] = P & \text{s.t.} \quad & \mathbb{E}_F\left[m(U;\theta,v)\right] = P \\
& \mathbb{E}_F\left[s(U;\theta,v)\right] \leq \bar{s} & & \mathbb{E}_F\left[s(U;\theta,v)\right] \leq \bar{s} \\
& \sup_{g\in\mathcal{G}} \mathbb{E}_F\left[\psi(U;\theta,v,g)\right] = 0 & & \sup_{g\in\mathcal{G}} \mathbb{E}_F\left[\psi(U;\theta,v,g)\right] = 0
\end{aligned} \tag{5}$$

where $\bar{s} \in \mathbb{R}$ is a user-specified threshold. The optimization problem searches for a distribution F in the identified set that results in $\mathbb{E}_F\left[s(U;\theta,v)\right] \leq \bar{s}$ and is the closest to the reference distribution F_0 in terms of KL divergence.

We can plot the bounds against δ and then find the radius corresponding to $\bar{s}$.

Remark 5. The δ^* in Section 5 can be obtained by removing the constraint for $\bar{s}$ in (5). That is, we seek the smallest radius δ^* such that the identified set is nonempty. See Schennach (2014) Page 356 and Christensen and Connault (2023) Section 3.3 for similar definitions.

6 Practical Implementation

This section presents the practical implementation of the proposed framework. Section 6.1 reviews the entropic optimal transport problem and the Sinkhorn algorithm. Section 6.2 proposes a computationally feasible algorithm.

6.1 Entropic Optimal Transport and Sinkhorn Algorithm

This section reviews the Sinkhorn algorithm for the entropic optimal transport problem.[21] Let $(\mathcal{U}_i, \nu_i)$ for $i = 1, \ldots, k$ be probability spaces, where $\mathcal{U}_i$ is the support for the random variable U_i. For a cost function $c : \mathcal{U}_1 \times \cdots \times \mathcal{U}_k \to \mathbb{R}$, the entropic optimal transport problem[22] with regularization parameter $\lambda_{KL} > 0$ is defined as:

$$\mathcal{C}_{\lambda_{KL}} := \inf_{F\in\Pi(\nu_1,\ldots,\nu_k)} \mathbb{E}_F\left[c(U_1, \ldots, U_k)\right] + \lambda_{KL} D_{KL}(F\|F_{\otimes}) \tag{EOT}$$

[21] See Sinkhorn and Knopp (1967), Cuturi (2013) and Nutz (2021).

[22] If $F_0 \neq F_{\otimes}$, then Lemma 13 reformulates the problem as the EOT problem.

whose dual is given by:

$$\mathcal{C}_{\lambda_{KL}} = \sup_{\{\phi_i \in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i}\phi_i(U_i) - \lambda_{KL}\mathbb{E}_{F_\otimes} \exp\left(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U_1, \ldots, U_k)}{\lambda_{KL}}\right) + \lambda_{KL} \quad (6)$$

where ϕ_i is the test function for the marginal distribution constraint ν_i. The dual problem is a concave maximization problem over $\{\phi_i\}_{i=1}^k$. The worst-case distribution is given by:

$$\frac{dF^*(U)}{dF_\otimes(U)} = \exp\left(\frac{\sum_{i=1}^k \phi_i^*(U_i) - c(U_1, \ldots, U_k)}{\lambda_{KL}}\right)$$

where the optimizers $(\phi_1^*, \ldots, \phi_k^*)$ are known as the optimal EOT potentials (also called Schrödinger potentials), which are the solutions to the Schrödinger equation (SE):

$$\phi_1(U_1) = -\lambda_{KL} \log\left(\mathbb{E}_{F_{\otimes,-1}} \exp\left(\frac{\sum_{i=2}^k \phi_i(U_i) - c(U_1, \ldots, U_k)}{\lambda_{KL}}\right)\right) \quad \nu_1\text{-a.s.} \quad \text{(SE1)}$$

$$\vdots$$

$$\phi_k(U_k) = -\lambda_{KL} \log\left(\mathbb{E}_{F_{\otimes,-k}} \exp\left(\frac{\sum_{i=1}^{k-1} \phi_i(U_i) - c(U_1, \ldots, U_k)}{\lambda_{KL}}\right)\right) \quad \nu_k\text{-a.s.} \quad \text{(SEk)}$$

where $F_{\otimes,-i}$ is the product measure of all marginals except for the i-th marginal. The Schrödinger equations (SE1) to (SEk) can be interpreted as the variational first-order conditions for optimality (see Nutz, 2021, Remark 3.4). Moreover, they also characterize the marginal constraints. To see this, define:

$$dF(U) = \exp\left(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U_1, \ldots, U_k)}{\lambda_{KL}}\right) dF_\otimes(U)$$

The marginal density can be obtained by integrating out the other marginals; therefore:

$$\text{(SEi)} \Leftrightarrow \text{the } i\text{-th marginal of } F \text{ is } \nu_i$$

The Sinkhorn algorithm can be interpreted as a coordinate ascent scheme for the optimization problem (6). It is a computationally fast[23], iterative method for solving (SE1)-(SEk).

Algorithm 1 (Sinkhorn Algorithm)**.** Initialize $\phi_i^{(0)} := 0$ for $i = 1, \ldots, k$. For iteration t,

[23] For its convergence rate, see Peyré et al. (2019), Carlier (2022), and Eckstein and Nutz (2022).

update ϕ_j for $j = 1, \ldots, k$ by:

$$\phi_j^{(t+1)}(U_j) := -\lambda_{KL} \log \int \exp\left(\frac{\sum_{i \neq j} \phi_i^{(t)}(U_i) - c(U_1, \ldots, U_k)}{\lambda_{KL}}\right) dF_{\otimes,-j}.$$

Stop if $\sup_j \|\phi_j^{(t+1)} - \phi_j^{(t)}\|_2 < \epsilon$ for a tolerance $\epsilon > 0$.

6.2 Proposed Algorithm

For a given $(\theta, v, g, \lambda, \lambda_{KL})$, the EOT problem provides the worst-case distribution, F^*. This allows us to update v by solving the structural constraint with F^*. We therefore propose an iterative algorithm to solve the minimax problem. The algorithm proceeds by alternating between updating (θ, v) and the dual (Lagrange multiplier) variables, $(g, \lambda, \lambda_{KL})$. After initializing all parameters, each iteration t involves the following steps:

Algorithm 2. Initialize $(\theta^{(0)}, v^{(0)}, g^{(0)}, \lambda^{(0)}, \lambda_{KL}^{(0)})$. At iteration t,

1. **Update Model Primitives** (θ, v)**:** For $(\theta^{(t)}, v^{(t)}, g^{(t)}, \lambda^{(t)}, \lambda_{KL}^{(t)})$, update[24] the model parameters (θ, v) by:

   (a) Propose a new candidate $\theta^{(t+1)}$.

   (b) Solve the EOT problem with $(\theta^{(t+1)}, v^{(t)}, g^{(t)}, \lambda^{(t)}, \lambda_{KL}^{(t)})$ and obtain F^*.

   (c) Update $v^{(t+1)}$ by solving the structural constraint with F^*.

   (d) Accept/reject the proposed $(\theta^{(t+1)}, v^{(t+1)})$.

2. **Update Dual Variables** $(g, \lambda, \lambda_{KL})$**:** For $(\theta^{(t+1)}, v^{(t+1)}, g^{(t)}, \lambda^{(t)}, \lambda_{KL}^{(t)})$, update $(g, \lambda, \lambda_{KL})$ following the same procedure as in the previous step.

3. Iterate until convergence, or a pre-specified number of iterations is reached.

The computational cost per iteration mainly comes from solving the EOT problem and the structural constraint, which are both computationally fast. However, the number of iterations required for convergence can be much larger, as our optimization problem is a minimax problem that is potentially non-differentiable.

[24] If gradient-based methods are used, then we smooth the non-differentiable components (e.g., the indicator function in Example 4) using a smooth approximation.

7 Empirical Application

This section applies our framework to an infinite-horizon dynamic demand model for new cars in the UK, France, and Germany. Due to the unobserved product characteristics, the indirect utility of purchasing is the latent variable. To estimate the price elasticity and conduct welfare analysis of an electric vehicle (EV) subsidy, we require a distributional assumption for the latent variable to solve the Bellman equation. Existing literature often uses an AR(1) process (e.g., Schiraldi, 2011; Gowrisankaran and Rysman, 2012), which may be misspecified. For example, the indirect utility may exhibit nonlinear dynamics. Therefore, we conduct a sensitivity analysis with respect to this reference distribution.

7.1 Data

We use the trim-level[25] data from IHS Markit during the period from 2014 to 2023. The monthly level dataset contains sales, list price, and characteristics of car models in the UK, France and Germany, which are treated as three independent markets in our analysis. To construct the final dataset, we first aggregate data from the trim-level to the model-level. Then, we aggregate fuel types into: petrol, diesel, electric, and hybrid. We remove models whose total sales during the data period are less than 20,000.[26] Finally, we adjust list prices by subtracting EV subsidies. The initial market size for January 2014 is calculated by subtracting the number of registered cars from the total population of each country. The market size is then updated each subsequent month by subtracting the total number of cars sold in the preceding period.

Table 1 presents the summary statistics. The three markets offer around 141–215 products from around 23 to 30 brands per month. In terms of average sales per model, Germany has 81,401 units, closely followed by France with 81,357 units, and the UK with 70,682 units. The average price is around \$33,444 in the UK, \$27,830 in France, and \$36,117 in Germany.

7.2 The Model

The model is infinite horizon. At each month t, a consumer i chooses $j \in \mathcal{J}_t \bigcup \{0\}$ where $\mathcal{J}_t$ is the set of available cars at t, and 0 is the outside option of not purchasing. Each car $j \in \mathcal{J}_t$

[25] In the automobile industry, a trim-level refers to a specific version of a vehicle model that comes with a particular set of features, options, and styling elements.

[26] In addition, we exclude car-month observations with sales below 150 units in the German market and below 5 units in the French market.

Table 1: Summary Statistics by Country (Monthly, 2014-2023)

Country	**Avg #** **Products**	**Avg #** **Brands**	**Avg #** **Sales**	**Price ($)** **Mean**	**Horsepower** **Mean**	**Weight (kg)** **Mean**
UK	196	30	70,682	33,444	140	1,836
France	141	23	81,357	27,830	112	1,702
Germany	215	26	81,401	36,117	151	1,952

Note: The price is adjusted for EV subsidies. First two columns are average number of products and brands per month. Average sales is the average number of cars sold per model. Mean price, horsepower, and weight are weighted by total sales.

is characterized by a vector of observable characteristics x_{jt}, price p_{jt}, and an unobserved characteristic ξ_{jt}. The period utility of choosing j is given by:

$$u(j, x_t, p_t, \xi_t, \varepsilon_{it}) = \begin{cases} \alpha p_{jt} + x_{jt}^T \theta + \xi_{jt} + \varepsilon_{ijt} & \text{if } j \in \mathcal{J}_t \\ \varepsilon_{0it} & \text{if } j = 0 \end{cases}$$

where ε_{it} is a vector of i.i.d. type I extreme value utility shocks, and x_t, p_t, ξ_t are the vectors of observable characteristics, prices, and unobserved characteristics for all cars in $\mathcal{J}_t$.

We assume a purchase is a terminating decision, i.e., consumers exit the market after the purchase. The conditional value function of purchasing car j can be written as the sum of the current period utility and the flow utility after purchase:

$$v_j(x_t, p_t, \xi_t) = \frac{x_{jt}^T \theta + \xi_{jt}}{1 - \beta} + \alpha p_{jt}$$

where $\beta = 0.975$ is the discount factor. The inclusive value of purchasing is defined as:

$$\omega_t = \log \sum_{j \in \mathcal{J}_t} \exp \left(\frac{x_{jt}^T \theta + \xi_{jt}}{1 - \beta} + \alpha p_{jt} \right)$$

Following Schiraldi (2011) and Gowrisankaran and Rysman (2012), we assume:

Assumption 10 (Inclusive Value Sufficiency[27] (IVS))**.** *$G(\omega_{t+1}|x_t, p_t, \xi_t)$ can be summarized by $G(\omega_{t+1}|\omega_t)$ where G is the conditional distribution function.*

Under the IVS assumption, ω_t is the only state variable, and the value function $V(\omega)$ is the solution to the smoothed Bellman equation:

$$V(\omega) = \log \left(\exp \left(v_0(\omega) \right) + \exp \left(v_1(\omega) \right) \right) \tag{7}$$

[27]The IVS assumption has also been used in Hendel and Nevo (2006); Melnikov (2013); Osborne (2018).

where $v_0(\omega) = \beta\mathbb{E}[V(\omega')|\omega]$ and $v_1(\omega) = \omega$ are the conditional value functions of not purchasing and purchasing, respectively. The market share of car j at time t is given by:

$$s_{jt}(x_t, p_t, \xi_t) = \underbrace{\frac{\exp(\omega_t)}{\exp(V(\omega_t))}}_{\text{Probability of Purchasing a Car}} \times \underbrace{\frac{\exp(v_j(x_t, p_t, \xi_t))}{\exp(\omega_t)}}_{\text{Conditional Probability of Purchasing Car } j} \tag{8}$$

7.3 First-Stage Estimation

In each market, the car with the highest total sales is set as the reference product, denoted by r.[28] Taking the log-odds ratio for cars j and r at time t yields:

$$\log\left(\frac{s_{jt}}{s_{rt}}\right) = \alpha\Delta p_{jt} + \frac{\Delta x_{jt}^T\theta}{1-\beta} + \frac{\Delta\xi_{jt}}{1-\beta} \tag{9}$$

where $\Delta p_{jt} := p_{jt} - p_{rt}$, $\Delta x_{jt} := x_{jt} - x_{rt}$, and $\Delta\xi_{jt} := \xi_{jt} - \xi_{rt}$. The parameters (α, θ) are identified[29] by the BLP instruments Berry et al. (1993). Moreover, $\Delta\xi_{jt}$ can be recovered by fitting the relative market share $\frac{s_{jt}}{s_{rt}}$, while the unobserved characteristic of the reference car, ξ_{rt}, cannot.

The exogenous characteristics, x_{jt}, include vehicle log-weight, log-horsepower, brand fixed effects, SUV fixed effect, and fuel type fixed effects.[30] Table 2 presents the regression results. Price coefficients are negative and significant in all markets, ranging from -0.158 in France to -0.192 in Germany. Relative to petrol vehicles, EVs are preferred in the UK (0.019) and France (0.004), but not in Germany (0.000). Hybrid vehicles are valued positively across all three markets, with coefficients ranging from 0.029 in Germany to 0.037 in France. In contrast, diesel has a negative coefficient in the UK (-0.015) but positive effects in Germany (0.008) and France (0.006).

[28] The reference cars are Volkswagen Golf (Petrol) in Germany, Peugeot 208 (Petrol) in France, and Ford Fiesta (Petrol) in the UK. The reference car for each country is always available in that country's market.

[29] The intercept and reference fixed effects are not identified from (9); they are absorbed into ξ_{rt}.

[30] The instruments are the exogenous product characteristics, average log-weight and log-horsepower of competitors' products, the proportion of competitors' products, the proportion of hybrid cars squared, and the number of brands. A competitor product is defined as a car whose brand is not that of r or j. The Kleibergen-Paap statistic for the first-stage regression is 17.93 for the UK, 33.76 for France, and 731.6 for Germany.

Table 2: Instrumental Variable Regression Results

	UK		Germany		France	
	Coef.	**Std. Err.**	**Coef.**	**Std. Err.**	**Coef.**	**Std. Err.**
Price	−0.178	(0.043)	−0.192	(0.004)	−0.158	(0.042)
Log horsepower	0.140	(0.038)	0.165	(0.003)	0.065	(0.031)
Log weight	−0.008	(0.003)	0.046	(0.003)	0.002	(0.001)
SUV	0.018	(0.002)	−0.003	(0.001)	0.013	(0.002)
Diesel	−0.015	(0.003)	0.008	(0.001)	0.006	(0.004)
Electric	0.019	(0.004)	0.000	(0.001)	0.004	(0.002)
Hybrid	0.035	(0.010)	0.029	(0.001)	0.037	(0.009)
Adjusted R^2	0.549		0.533		0.803	
# of Month-Years	120		120		120	
# of Obs.	23,451		25,715		16,862	

Note: An observation is a pair of (j, t) where j is a car model other than the reference car r and t is the time period. Standard errors are in parentheses. Brand fixed effects are not reported.

7.4 The Reference Distribution and Scalar Parameters of Interest

We first define the reference distribution and then introduce the scalar parameters of interest. After the first stage estimation, we can calculate ω_t up to ξ_{rt} as:

$$\omega_t = \log \sum_{j \in \mathcal{J}_t} \exp\left(\frac{x_{jt}^T\theta + \Delta\xi_{jt}}{1-\beta} + \alpha p_{jt}\right) + \frac{\xi_{rt}}{1-\beta} \tag{10}$$

As we have identified the utility parameters, the potential sensitivity of the empirical results solely arises from the distributional assumption on ω_t.

The reference transition of ω_t to solve the Bellman equation is an AR(1) process:

$$\omega_t = \gamma_0 + \gamma_1\omega_{t-1} + \eta_t \tag{11}$$

where η_t follows an i.i.d. normal distribution with mean 0 and variance σ^2.

The parameters $(\gamma_0, \gamma_1, \sigma^2)$ are estimated using an iterative procedure. We begin with an initial guess of $(\gamma_0, \gamma_1, \sigma^2)$ and circulate between: (i) solving the Bellman equation (7), (ii) recovering $\{\omega_t\}_{t=1}^T$ from the market share of purchasing (the first part of (8)), (iii) updating $(\gamma_0, \gamma_1, \sigma^2)$ by refitting an AR(1) process (11) until we find a fixed point. The reference distribution F_0 for (ω, ω') is the product of the transition kernel of the estimated AR(1) process and its stationary distribution ν_0. The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F\|F_0) \leq \delta\}$$

We consider two scalar parameters: (i) the average industrywide price elasticity of demand, (ii) the welfare analysis of an additional EV subsidy. For the second counterfactual, we focus on an additional \$3,000 EV subsidy in 2023, which is the last year of our data. This counterfactual is motivated by recent policy changes in our sample markets: the UK plug-in car grant was withdrawn in June 2022, Germany's Umweltbonus was terminated in December 2023, and France's bonus écologique was tightened in late 2023 through updated eligibility criteria. The amount is also of the same order of magnitude as historical EV purchase subsidies in these countries, including the €3,000–€4,500 German Umweltbonus in 2023, the €5,000 French bonus écologique in 2023 (with an additional €2,000 top-up for lower-income households), and the £1,500 UK plug-in car grant before its withdrawal. For both cases, the transition of ω_t is unchanged, i.e., we assume consumers' beliefs about the transition of ω_t stay the same.

For the average industrywide price elasticity, we consider a 1% increase in the price of all cars. The future ω_{t+1} is conditional on:

$$\omega_t' = \log \sum_{j \in \mathcal{J}_t} \exp\left(\frac{x_{jt}^T\theta + \Delta\xi_{jt}}{1-\beta} + 1.01 \cdot \alpha p_{jt}\right) + \frac{\xi_{rt}}{1-\beta}$$

and the average industrywide price elasticity is:

$$\frac{1}{T}\sum_{t=1}^{T} \frac{s_{0t} - s_0(\omega_t')}{1 - s_{0t}} \times 100$$

where $s_0(\omega_t')$ is the model-implied market share of not purchasing.

For the EV subsidy, denote by $\mathcal{J}_{t,\text{EV}}$ the set of EVs at t. The future ω_{t+1} is conditional on:

$$\omega_t^{\text{EV}} = \log \sum_{j \in \mathcal{J}_t} \exp\left(\frac{x_{jt}^T\theta + \Delta\xi_{jt}}{1-\beta} + \alpha\left(p_{jt} - \mathbb{1}(j \text{ is an EV}) \cdot 3000\right)\right) + \frac{\xi_{rt}}{1-\beta}$$

and the consumer surplus from the subsidy is given by:

$$\text{Consumer Surplus} = \sum_{t=t_1}^{T} \frac{V(\omega_t^{\text{EV}}) - V(\omega_t)}{-\alpha} \times M_t$$

where M_t is the market size at time t and t_1 is January 2023.

7.5 Sensitivity Analysis

Our framework requires the constraints to be linear in F, while the Bellman equation (7) is not. We first reformulate it to fit into our framework. By the *Hotz-Miller Inversion Lemma* (Hotz and Miller, 1993), we have:

$$V(\omega) = \omega - \log s_1(\omega) \tag{12}$$

Taking the log-odds ratio of purchasing and not purchasing, and using (12), we have:

$$\log\left(\frac{1 - s_0(\omega)}{s_0(\omega)}\right) = \omega - \beta\mathbb{E}\left[\omega' - \log(1 - s_0(\omega'))|\omega\right] \tag{13}$$

The above constraint is the fixed point problem on the market share space (see Aguirregabiria and Mira, 2002). The right-hand side is linear in the conditional distribution. The following lemma establishes the relationship between the fixed point problems in (7) and (13).

Assumption 11. *Assume the inclusive value ω has compact support $\Omega \subset \mathbb{R}$ with nonempty interior equipped with the sup-norm and $\mathbb{E}\left[f(\omega')|\omega\right] \in C(\Omega)$ for any $f \in C(\Omega)$ where $C(\Omega)$ is the space of continuous functions on Ω.*

Lemma 3. *The following holds:*

(i) *Under Assumption 11, the fixed point problem* (7) *has a unique solution on* $C(\Omega)$.

(ii) *The fixed point problem* (13) *has a unique solution if and only if the fixed point problem* (7) *has a unique solution.*

(iii) *If* (7) *and* (13) *both have unique solutions, then it holds that* $1-s_0(\omega) = \exp\left(\omega - V(\omega)\right)$ *where* $V(\omega)$ *and* $s_0(\omega)$ *are the solutions to* (7) *and* (13), *respectively.*

Lemma 3 shows that solving the Bellman equation (7) is equivalent to solving (13). We further convert (13) into an unconditional moment constraint by assuming that $s_0(\omega)$ is the solution to (13) if and only if:

$$\sup_{g \in C(\Omega)} \mathbb{E}_F\left[g(\omega)\left(\log(\frac{1 - s_0(\omega)}{s_0(\omega)}) - \omega + \beta\omega' - \beta\log(1 - s_0(\omega'))\right)\right] = 0 \tag{14}$$

Assumption 12. *For $\forall\ F \in \mathcal{F}$, the solution $s_0(\omega)$ corresponding to* (13) *satisfies the following: for all $t = 1, \ldots, T$, there exists a unique $\omega_t \in \Omega$ such that $s_0(\omega_t) = s_{0t}$.*

Assumption 12 allows us to profile out $\{\omega_t\}_{t=1}^T$, which is useful for the implementation. A sufficient condition is that $s_0(\omega)$ is continuous and strictly decreasing, and its smallest and largest values are small and large enough. Under Assumption 11, we have $s_0(\omega) \in C(\Omega)$. Moreover, we can expect that a higher inclusive value ω corresponds to a lower market share of not purchasing, i.e., $s_0(\omega)$ is decreasing in ω. Therefore, Assumption 12 is mild.

The last condition is the fixed point constraint similar to the procedure to estimate the AR(1) process. Suppose the distribution F is used in (14), and $\{\omega_t\}_{t=1}^T$ is the sequence of recovered inclusive values. Denote by $\hat{F}$ the estimator of the joint distribution for the pairs $\{(\omega_t, \omega_{t+1})\}_{t=1}^{T-1}$. Then, our fixed point constraint is:

$$D_{KL}(F\|\hat{F}) \leq \epsilon_T$$

where ϵ_T is the tolerance level. To choose ϵ_T, we estimate the joint distribution of inclusive values recovered from the AR(1) process by the kernel density estimator with Gaussian kernel and bandwidth selected by the 5-fold cross-validation. Then, we set ϵ_T to be the KL divergence between the kernel density estimator and the reference distribution.

For EV subsidy, by (12), the consumer surplus (CS) is given by:

$$V(\omega_t^{\text{EV}}) - V(\omega_t) = \omega_t^{\text{EV}} - \omega_t - \log(1 + \frac{s_1(\omega_t^{\text{EV}}) - s_{1t}}{s_{1t}})$$

where $\omega_t^{\text{EV}} - \omega_t$ does not depend on F. Therefore, to bound CS, it is equivalent to bounding the change in purchase share.

Putting everything together, the lower bound on the average elasticity is given by:

$$\begin{aligned}
\inf_{s_0(\omega)\in C(\Omega)} \inf_{F\in\mathcal{F}} \quad & \frac{1}{T}\sum_{t=1}^{T} \frac{s_{0t} - s_0(\omega_t')}{1 - s_{0t}} \times 100 \\
\text{s.t.} \quad & s_0(\omega_t) = s_{0t} \text{ for } t = 1, \ldots, T \\
& \sup_{g\in C(\Omega)} \mathbb{E}_F \left[g(\omega) \left(\log(\frac{1 - s_0(\omega)}{s_0(\omega)}) - \omega + \beta\omega' - \beta \log(1 - s_0(\omega')) \right) \right] = 0 \\
& D_{KL}(F\|\hat{F}) \leq \epsilon_T
\end{aligned}$$

For $\delta = 0$, the reference distribution is the unique solution to the above problem. The corresponding elasticity is the reference industrywide elasticity.

7.6 Implementation

We adapt Algorithm 2 proposed in Section 6.2. We will use numerical integration to discretize the support of the AR(1) process. Therefore, the recovered inclusive values $\{\omega_t\}_{t=1}^T$ are not differentiable with respect to the discretized market share function $s_0(\omega)$. To address this issue, we employ the MCMC optimization method. We alternate between solving the EOT problem to obtain the worst-case distribution, solving the Bellman equation to update $s_0(\omega)$, recovering the inclusive values, and checking the fixed point constraint. To derive a tractable dual formulation, we handle the fixed point constraint in a specific way. Because this constraint depends on an estimator, $\hat{F}$, that changes during optimization—potentially causing numerical instability—we first derive the dual formulation without it.[31] Then, the fixed point constraint determines the acceptance/rejection of the candidate parameters in the Metropolis-Hastings step. Consider the following optimization problem:

$$\begin{aligned} \inf_{s_0(\omega)\in C(\Omega)} \inf_{F\in\mathcal{F}} \quad & \frac{1}{T}\sum_{t=1}^{T} \frac{s_{0t}-s_0(\omega_t')}{1-s_{0t}} \times 100 \\ \text{s.t.} \quad & s_0(\omega_t) = s_{0t} \text{ for } t=1,\dots,T \\ & \sup_{g\in C(\Omega)} \mathbb{E}_F\left[g(\omega)\left(\log(\frac{1-s_0(\omega)}{s_0(\omega)}) - \omega + \beta\omega' - \beta\log(1-s_0(\omega'))\right)\right] = 0 \end{aligned}$$

Applying Theorem 1, its dual is:

$$\begin{aligned} \inf_{s_0(\omega)\in C(\Omega)} \sup_{g\in C(\Omega),\lambda_{KL}\geq 0} & \frac{1}{T}\sum_{t=1}^{T} \frac{s_{0t}-s_0(\omega_t')}{1-s_{0t}} \times 100 + \mathcal{C}(s_0,g,\lambda_{KL}) - \lambda_{KL}\delta \\ \text{s.t.} \quad & s_0(\omega_t) = s_{0t} \text{ for } t=1,\dots,T \end{aligned}$$

where $\mathcal{C}(s_0,g,\lambda_{KL})$ is the EOT problem: $\mathcal{C}(s_0,g,\lambda_{KL}) := \inf_{F\in\Pi(\nu_0,\nu_0)} \mathbb{E}_F\left[c(\omega,\omega';s_0,g)\right] + \lambda_{KL}D_{KL}(F\|F_0)$ whose cost function is $c(\omega,\omega';s_0,g) = g(\omega)\left(\log(\frac{1-s_0(\omega)}{s_0(\omega)}) - \omega + \beta\omega' - \beta\log(1-s_0(\omega'))\right)$ and the worst-case conditional distribution F^* is given by:

$$dF^*(\omega'|\omega) = \exp(\frac{\phi_1^*(\omega)+\phi_2^*(\omega')-c(\omega,\omega';s_0,g)}{\lambda_{KL}})dF_0(\omega'|\omega) \quad F_0\text{-a.s.}$$

where $\phi_1^*(\omega)$ and $\phi_2^*(\omega')$ are the optimal EOT potentials. During the optimization process, $dF^*(\omega'|\omega)$ is used to update $s_0(\omega)$ by solving (13) using fixed point iteration.

As shown in Theorem 1, the expectation in the dual is taken with respect to the reference

[31] In principle, we can choose other constraints like the integral probability metrics and adapt the minimax theorem in Theorem 1. However, it can lead to more optimization parameters.

distribution. Therefore, we discretize the estimated AR(1) process, which results in three approximation errors: (i) the Bellman equation is solved on the discretized support, (ii) $\{\omega_t\}_{t=1}^T$ are recovered approximately, and (iii) the elasticity is computed approximately. That is, we approximate $s_0(\omega)$ by the market share of the nearest grid point to ω. Therefore, there is a trade-off between approximation error and computational cost. A finer discretization reduces the approximation error, while increasing the number of optimization parameters.

However, our dual formulation significantly improves computational efficiency. Suppose we discretize the AR(1) process into N grid points. If we directly solve (14), the number of optimization parameters is $O(N^2)$ due to the transition matrix. In contrast, the number of optimization parameters is $O(N)$ in the dual formulation.

Algorithm 3 summarizes the simulated annealing MCMC optimization algorithm (Kirkpatrick et al., 1983). It starts with the reference market share $s^{(0)}$ and alternates between proposing new parameters (g', λ'_{KL}), solving the EOT problem, solving the Bellman equation using the worst-case distribution, and accepting or rejecting the proposed parameters using the Metropolis-Hastings step based on the change in the elasticity penalized by the violation of the market share and fixed-point constraints. At each improvement step, it pools previous results across all radii for initialization. We choose 99 grid points, 5,000 MCMC steps, 5 optimization steps, 14 radii (the last is 10^{10}), and 100 as the simulated annealing multiplier.

Inference: We construct confidence intervals for the bounds using the numerical delta method of Hong and Li (2018). We fix the demand parameters at their point estimates, and only account for the estimation uncertainty of the AR(1) parameters, which are estimated by the fixed-point constraint.

Let $\theta_f := (\gamma_0, \gamma_1, \sigma)$ be the vector of AR(1) parameters, and let $\Psi_T(\theta_f)$ denote one iteration of the fixed-point mapping: construct the transition matrix at θ_f, solve the Bellman equation, recover inclusive values, and refit an AR(1). At convergence, $\hat{\theta}_f = \Psi_T(\hat{\theta}_f)$. Linearizing this fixed-point equation implies that the asymptotic covariance of $\sqrt{T}(\hat{\theta}_f - \theta_f)$ is $\hat{\Sigma} = (I - \hat{J})^{-1}\hat{V}(I - \hat{J})^{-\top}$, where $\hat{J}$ is the Jacobian of the fixed-point mapping evaluated at the fixed point, and $\hat{V}$ is the asymptotic covariance of the one-step AR(1) refit.

The procedure is as follows. Let $\varepsilon_T := T^{-1/3}$ be the perturbation step size, which satisfies $\varepsilon_T \to 0$ and $\varepsilon_T\sqrt{T} \to \infty$, and let $r_T := \sqrt{T}$ be the convergence rate. For $b = 1, \ldots, B$:

1. Draw $Z_b \sim \mathcal{N}(0, \hat{\Sigma})$ and set $\theta_{f,b} = \hat{\theta}_f + \varepsilon_T Z_b$.

2. At $\theta_{f,b}$, construct the transition matrix on the original grid, solve the Bellman equation, recover inclusive values from the original market shares, and re-run Algorithm 3 to

Algorithm 3: Simulated Annealing MCMC Optimization Algorithm

```
Parameters: N: Number of grid points; S: MCMC steps per optimization run; J: Optimization
            steps; m: Simulated annealing multiplier; l: Number of radii;
for j = 1 to J                                              /* Optimization Step j */
do
  for i = 0 to l − 1, set δ_i = 10^(−3+i·0.25)
  do
    if i = 0 then
      If j = 0: Set s^(0) as the reference market share. Initialize g^(0), λ_KL^(0).
      If j > 0: Set (s^(0), g^(0), λ_KL^(0)) to the optimal solution from the previous step's stored
        results with upper bound 10^(−3) on the KL divergence to the reference distribution.
    else
      If j = 0: Set (g^(0), λ_KL^(0), s^(0)) to the optimal solution from the previous step.
      If j > 0: Set (s^(0), g^(0), λ_KL^(0)) to the optimal solution from the previous stored results
        with upper bound 10^(−3+i·0.25) on the KL divergence to the reference distribution.
    end
    for s = 1, ..., S                          /* Simulated Annealing MCMC optimization */
    do
      // 1.  Propose New Parameters
      Propose (g', λ'_KL) from the random walk, solve the EOT problem C(s_0^s, g', λ'_KL) and
        obtain F*, solve the Bellman equation (13) with F*, recover {ω_t}_{t=1}^T, estimate the
        distribution of {(ω_t, ω_{t+1})}_{t=1}^T by kernel density estimator, and compute the elasticity.
      // 2.  Check Constraints and Apply Penalty
      Calculate the sum of violations from the market share and fixed-point constraints. The
        market share violation is defined as max{0, vio_F − vio_ref} where vio_ref is the
        violation of the reference model. If the total violation exceeds 0.005, add a large
        penalty (100). If D_KL(F*||F_0) > δ_i, add a large penalty (100).
      // 3.  Accept/Reject (Metropolis-Hastings)
      Apply a Metropolis-Hastings step based on the change in the (penalized) elasticity
        multiplied by 10*(1+(s−1)*(m−1))/(S−1) where the prior is N(0, 100).
      // 4.  Adapt (Andrieu and Thoms, 2008, Algorithm 4)
      Update the random walk via vanishing adaptation scheme.
    end
  end
end
```

compute $\hat{\phi}_b(\delta)$, the bound at the perturbed parameters, for each radius δ. Note that we choose 10,000 MCMC steps and 1 optimization step for the re-run.

3. Compute the numerical derivative:

$$D_b^*(\delta) = \frac{\hat{\phi}_b(\delta) - \hat{\phi}(\delta)}{\varepsilon_T}$$

where $\hat{\phi}(\delta)$ is the point estimate from the original (unperturbed) optimization.

We discard draws for which the perturbed AR(1) parameters violate the stationarity condition ($|\gamma_{1,b}| \geq 1$). Denoting the retained draws by $\{D_b^*(\delta)\}_{b\in\mathcal{B}}$, the 95% confidence interval for $\hat{\phi}(\delta)$ is:

$$\left[\hat{\phi}(\delta) - \frac{c_{0.95}(\delta)}{r_T}, \quad \hat{\phi}(\delta) + \frac{c_{0.95}(\delta)}{r_T}\right]$$

where $c_{0.95}(\delta)$ is the 95th percentile of $\{|D_b^*(\delta)|\}_{b\in\mathcal{B}}$. For bounds where only the one-sided direction matters (e.g., lower bound), we report the one-sided confidence band. We set $B = 500$ and the confidence level to 95%.

7.7 Results

We estimate two alternative transition densities for each market. The first assumes that the inclusive values are i.i.d. normally distributed. The second estimates a nonlinear AR(1) process using a cubic spline[32]. In the following figures, we plot the KL divergence from the alternative models to the reference model. The independent model is closer to the reference model with KL divergence between 0.04 and 0.67, while the nonlinear AR(1) process is farther away, with KL divergence between 3.94 and 10.09.

Figures 1-3 plot the bounds on the average industrywide price elasticity in 2023 for the UK, Germany, and France.[33] France is the least elastic, while Germany is the most elastic. At $\delta = 0.001$, the bounds are $[-5.765, -5.675]$ for the UK, $[-6.360, -6.274]$ for Germany, and $[-4.143, -4.110]$ for France. As δ increases, the bounds widen and eventually flatten. At $\delta = 10^{10}$, they are $[-5.871, -5.588]$ for the UK, $[-6.459, -6.236]$ for Germany, and $[-4.185, -4.078]$ for France. To interpret these results, we use the three sensitivity measures in Section 5. The local (global) sensitivity is the ratio of the interval length at $\delta = 0.001$ ($\delta = 10^{10}$) to the reference elasticity, while the robustness metric is the smallest KL radius at which a bound crosses the $\pm 2.5\%$ threshold around the reference value. The local/global sensitivity measures are 1.57%/4.94% for the UK, 1.37%/3.54% for Germany, and 0.81%/2.60% for France. Hence, the average elasticity is relatively robust in all three markets. Moreover, none of the elasticity bounds cross the $\pm 2.5\%$ threshold.

[32] The nonlinear AR(1) process is specified as: $g(\omega) = \sum_{k=1}^{N+3} \rho_k \Phi\left(\frac{\omega - a}{h} - (k-2)\right)$ where a is the minimum of the discretized support of ω, h is the distance between two adjacent grid points, ρ_k are parameters to be estimated, and $\Phi(t) = \begin{cases} 4 - 6t^2 + 3|t|^3 & \text{if } |t| \le 1 \\ (2-|t|)^3 & \text{if } 1 < |t| \le 2 \\ 0 & \text{otherwise} \end{cases}$. For this model, we set $N = 4$ to avoid overfitting. Then, we discretize the reference AR(1) process into 4 grid points, and compute the KL divergence from the nonlinear AR(1) process to the reference AR(1) process.

[33] Schiraldi (2011) finds a average long-run price elasticities ranging from -3.54 to -4.34 across different car segments for the Italian market. D'Haultfœuille et al. (2019) reports average elasticities between -3.94 and -6.40 across consumer groups for the French market. Reynaert and Sallee (2021) finds a mean own-price elasticity of -5.45 for the European market. Grieco et al. (2024) estimates an average elasticity of -5.36 for the U.S. market in 2015. Remmy (2025) reports a mean price elasticity of -4.043 for the German market.

Figures 1-3 also plot the bounds on consumer surplus in 2023 from an additional \$3,000 EV subsidy. The lower bounds remain positive in all three markets across all radii, implying that the subsidy is welfare-improving under all perturbations we consider. At $\delta = 0.001$, the bounds are [\$766 million,\$815 million] for the UK, [\$389 million,\$522 million] for Germany, and [\$852 million,\$945 million] for France. At $\delta = 10^{10}$, they widen to [\$537 million,\$1,185 million] for the UK, [\$216 million,\$602 million] for Germany, and [\$657 million,\$1,121 million] for France. For consumer surplus, we define local and global sensitivity in the same way, and define the robustness metric as the smallest KL radius at which a bound crosses the $\pm 10\%$ threshold around the reference consumer surplus. The local/global sensitivity measures are 6.19%/82.27% for the UK, 29.60%/85.61% for Germany, and 10.35%/51.52% for France. Thus, consumer surplus is substantially more sensitive than elasticity. The UK market is the least sensitive locally, France is the least sensitive globally, and Germany is the most sensitive across all three measures. The robustness metric for a 10% deviation from the reference consumer surplus is below 0.001 in Germany, around 0.006 in France, and around 0.018 in the UK.

The shaded confidence bands show that the misspecification error is quantitatively important. In both the elasticity and consumer surplus exercises, the sampling error around each bound is often of the same order as the misspecification-induced interval length, especially at small radii. Therefore, the empirical conclusions should be interpreted as reflecting two comparable sources of uncertainty: misspecification of the latent-state dynamics and sampling variation in the estimated reference process.

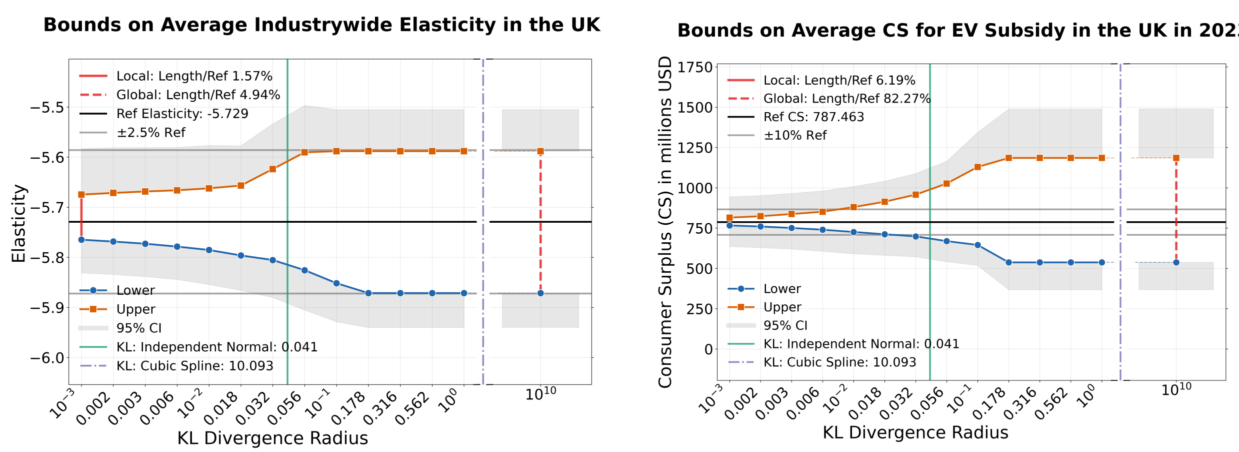

Figure 1: Bounds on Industrywide Elasticity and Consumer Surplus for the UK

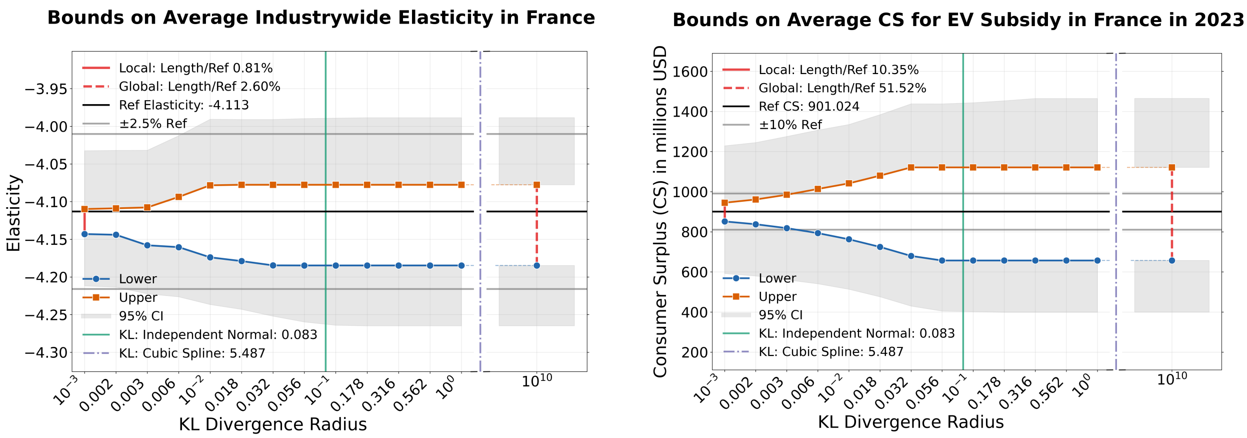

Figure 2: Bounds on Industrywide Elasticity and Consumer Surplus for France

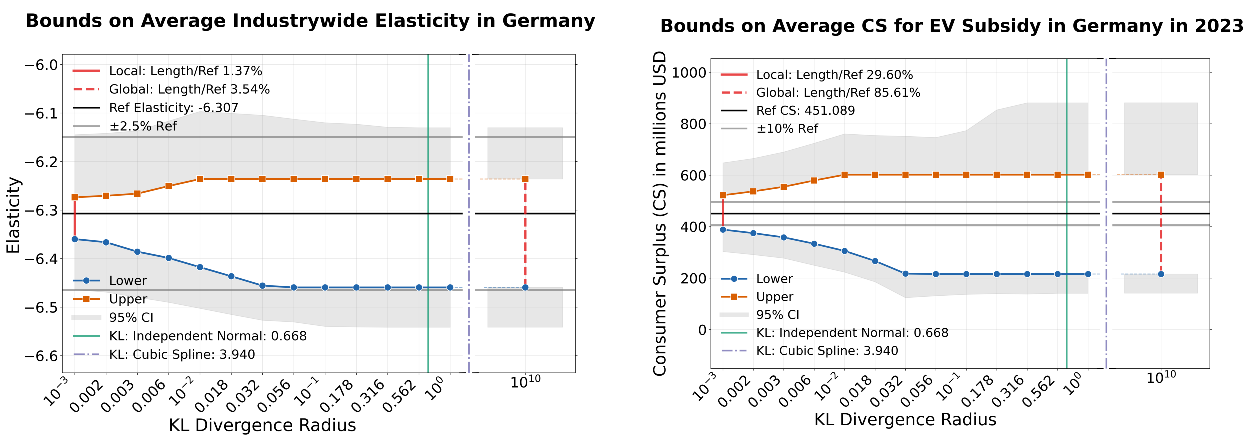

Figure 3: Bounds on Industrywide Elasticity and Consumer Surplus for Germany

8 Conclusion

We propose a computationally tractable framework to quantify the sensitivity of outcomes of interest to misspecified latent-state dynamics in structural models. We derive bounds on a scalar parameter of interest by perturbing a reference dynamic process, while imposing a stationarity condition for time-homogeneous models or a Markovian condition for time-inhomogeneous models. We apply the approach to an infinite-horizon dynamic demand model for new cars in the UK, Germany, and France.

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A Additional Examples and Results

Example 4 (Infinite Horizon Dynamic Discrete Choice Models)**.** This example considers the serial independence assumption on utility shocks in a single-agent DDC model, as in Rust (1987). Agents solve the Bellman equation for the conditional value function $v \in \mathcal{V}$:

$$v_j(x,\varepsilon) = u_j(x,\varepsilon;\theta) + \beta \mathbb{E}_{\varepsilon'|\varepsilon} \mathbb{E}_{x'|x,j} \max_{j' \in \mathcal{J}} v_{j'}(x',\varepsilon') \tag{15}$$

where $\varepsilon \in \mathbb{R}^J$ is a vector of utility shocks for each action $j \in \mathcal{J}$, $x \in \mathcal{X}$ is the observable state variable, $\beta \in (0,1)$ is the discount factor, $u_j(x,\varepsilon;\theta)$ is the period utility parameterized by $\theta \in \Theta$, and $\mathcal{V}$ is the class of conditional value functions.

Let $U := (\varepsilon, \varepsilon')$ be a vector of current and future utility shocks. The serial independence assumption is often imposed on utility shocks, which implies a reference distribution:

$$dF_0(U) := \nu_0(d\varepsilon)\nu_0(d\varepsilon')$$

whose perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0,\nu_0), D_{KL}(F\|F_0) \leq \delta\}$$

Suppose the scalar parameter of interest is the social welfare, defined as:

$$\mathbb{E}_{\nu_0} \mathbb{E}_x \max_{j \in \mathcal{J}} v_j(x,\varepsilon)$$

We convert the Bellman equation (15) into a restriction that depends on the joint distribution $F \in \mathcal{F}$. We assume there exists a class of test functions $\mathcal{G}$, chosen independently of F, such that for any $F \in \mathcal{F}$, v solves the Bellman equation (15) if and only if:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,j,x'} \left[g_j(x,\varepsilon) \left(v_j(x,\varepsilon) - u_j(x,\varepsilon;\theta) - \beta \max_{j' \in \mathcal{J}} v_{j'}(x',\varepsilon') \right) \right] = 0$$

The class $\mathcal{G}$ is assumed to be rich enough under every admissible F so that the unconditional restriction is equivalent to the corresponding conditional structural equation. We can rewrite the structural constraint as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \left[\psi(U;\theta,v,g) \right] = 0$$

where $\psi(U;\theta,v,g) := \mathbb{E}_{x,j,x'} \left(g_j(x,\varepsilon) \left(v_j(x,\varepsilon) - u_j(x,\varepsilon;\theta) - \beta \max_{j' \in \mathcal{J}} v_{j'}(x',\varepsilon') \right) \right)$.

We consider the following moment conditions for estimation:

$$\mathbb{E}_F \mathbb{1}(v_j(x,\varepsilon) = \max_{j'\in\mathcal{J}} v_{j'}(x,\varepsilon)) = P_0(j|x) \quad \forall\ (j,x) \in \mathcal{J}\times\mathcal{X}$$

where $\mathbb{1}$ is the indicator function, and $P_0(j|x)$ is the population CCP. We assume $\mathcal{X}$ has discrete support, and rewrite the moment conditions as:

$$\mathbb{E}_F\left[m(U;v)\right] = P_0$$

where $m(U;v)$ stacks the indicator functions for each (x,j) given v, and P_0 stacks the CCPs.

Then, the lower bound on the social welfare is given by:

$$\begin{aligned} \inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} & \mathbb{E}_{\nu_0}\mathbb{E}_x \max_{j\in\mathcal{J}} v_j(x,\varepsilon) \\ \text{s.t.} \quad & \mathbb{E}_F\left[m(U;v)\right] = P_0 \\ & \sup_{g\in\mathcal{G}} \mathbb{E}_F\left[\psi(U;\theta,v,g)\right] = 0 \end{aligned}$$

Example 5 (Infinite Horizon Dynamic Discrete-Continuous Choice Models)**.** This example considers the serial independence assumption on the consumption shock in a single-agent dynamic discrete-continuous choice model, as in Iskhakov et al. (2017) and Levy and Schiraldi (2022). At each period, individuals make a discrete choice $j\in\mathcal{J}$, and a continuous choice $q\in\mathbb{R}$. The value function solves the Bellman equation:

$$V(x,I,\xi,\varepsilon) = \max_{j,q}\left\{u_j(q,x,I,\xi;\theta) + \varepsilon_j + \beta\mathbb{E}_{\xi'|\xi}\mathbb{E}_{x'|x,j}V(x',I',\xi',\varepsilon')\right\}$$

where I is the resource constraint (e.g., inventory) evolving deterministically over time according to $I' = L(x,I,q,j)$, $x\in\mathcal{X}$ is the observable state variable, $\varepsilon\in\mathbb{R}^J$ is a vector of i.i.d. Extreme Value Type I utility shocks, $\xi\in\Xi$ is the consumption shock, $u_j(q,x,I,\xi;\theta)$ is the period utility parameterized by $\theta\in\Theta$, and $\beta\in(0,1)$ is the discount factor.

Let $q_j^* := q_j^*(x,I,\xi)$ be the conditional optimal continuous choice for (j,x,I,ξ). Given q_j^*, the conditional value function $v_j\in\mathcal{V}$ solves:

$$v_j(x,I,\xi) = u_j(q_j^*,x,I,\xi;\theta) + \beta\mathbb{E}_{\xi'|\xi}\mathbb{E}_{x'|x,j}\left[\log\left(\sum_{j'\in\mathcal{J}}\exp\left(v_{j'}(x',I',\xi')\right)\right)\right] + \beta\gamma \tag{16}$$

The model-implied conditional choice probability is $p(j|x,I,\xi) = \frac{\exp(v_j(x,I,\xi))}{\sum_{j'\in\mathcal{J}}\exp\left(v_{j'}(x,I,\xi)\right)}$.

We assume that the optimal continuous choice is attained at an interior point and the dominated convergence theorem holds. Therefore, the first-order condition for (16) holds:

$$\frac{\partial u_j(q_j^*, x, I, \xi; \theta)}{\partial q} + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x,j} \left[\sum_{j' \in \mathcal{J}} p(j'|x', I', \xi') \frac{\partial v_{j'}(x', I', \xi')}{\partial I'} \right] \frac{\partial L(x, I, q_j^*, j)}{\partial q} = 0$$

By the envelope theorem, we have:

$$\frac{\partial v_{j'}(x', I', \xi')}{\partial I'} = \frac{\partial u_{j'}(q_{j'}^*, x', I', \xi'; \theta)}{\partial q}$$

Therefore, the Euler equation for the optimal continuous choice is:

$$\frac{\partial u_j(q_j^*, x, I, \xi; \theta)}{\partial q} + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x,j} \left[\sum_{j' \in \mathcal{J}} p(j'|x', I', \xi') \frac{\partial u_{j'}(q_{j'}^*, x', I', \xi'; \theta)}{\partial q} \right] \frac{\partial L(x, I, q_j^*, j)}{\partial q} = 0 \tag{17}$$

Let $U := (\xi, \xi')$ be a vector of current and future consumption shocks. The serial independence is often imposed on consumption shocks, which implies a reference distribution:

$$dF_0(U) := \nu_0(d\xi)\nu_0(d\xi')$$

whose perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

Suppose the scalar parameter of interest is the social welfare, defined as:

$$\mathbb{E}_{\nu_0} \mathbb{E}_{x,I} \left[\log \left(\sum_{j \in \mathcal{J}} \exp\left(v_j(x, I, \xi)\right) \right) \right]$$

We convert the Bellman equation (16) and the Euler equation (17) into restrictions that depend on the joint distribution $F \in \mathcal{F}$. We assume there exists a class of test functions $\mathcal{G}$, chosen independently of F, such that for any $F \in \mathcal{F}$, $v := (v_1, \cdots, v_J, q_1^*, \cdots, q_J^*)$ solves the Bellman equation (16), and the Euler equation (17) if and only if:

$$\sup_{g^1 \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,I,j,x'} \left[g_j^1(x, I, \xi) \left(v_j(x, I, \xi) - u_j(q_j^*, x, I, \xi; \theta) - \beta \log \left(\sum_{j' \in \mathcal{J}} \exp\left(v_{j'}(x', I', \xi')\right) \right) - \beta\gamma \right) \right] = 0$$

$$\sup_{g^2 \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,I,j,x'} \left[g_j^2(x, I, \xi) \left(\frac{\partial u_j(q_j^*, x, I, \xi; \theta)}{\partial q} + \beta \sum_{j' \in \mathcal{J}} p(j'|x', I', \xi') \frac{\partial u_{j'}(q_{j'}^*, x', I', \xi'; \theta)}{\partial q} \frac{\partial I'(x, I, q_j^*, j)}{\partial q} \right) \right] = 0$$

The class $\mathcal{G}$ is assumed to be rich enough under every admissible F so that the unconditional restriction is equivalent to the corresponding conditional structural equation. We rewrite the structural constraints as $\sup_{g\in\mathcal{G}} \mathbb{E}_F [\psi(U;\theta,v,g)] = 0$ where ψ is the sum of the two expressions inside the expectations and $g := (g^1, g^2)$.

We consider the following moment conditions for estimation: $\forall\ (j,x,I) \in \mathcal{J} \times \mathcal{X} \times \mathcal{I}$

$$\mathbb{E}_F [p(j|x,I,\xi)] = P_0(j|x,I), \quad \mathbb{E}_F \left[q_j^*(x,I,\xi)\right] = q_0(j,x,I)$$

where $P_0(j|x,I)$ is the population CCP and $q_0(j,x,I)$ is the population continuous choice function. We discretize $\mathcal{X}\times\mathcal{I}$ for estimation, and rewrite the moment conditions as $\mathbb{E}_F [m(U;v)] = P_0$ where m (P_0) stacks the model-implied (population) CCPs and continuous choice functions.

Then, the lower bound on the social welfare is given by:

$$\begin{aligned}
\inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} \ & \mathbb{E}_{\nu_0}\mathbb{E}_{x,I} \log\left(\sum_{j\in\mathcal{J}} \exp\left(v_j(x,I,\xi)\right)\right) \\
\text{s.t.} \quad & \mathbb{E}_F [m(U;v)] = P_0 \\
& \sup_{g\in\mathcal{G}} \mathbb{E}_F [\psi(U;\theta,v,g)] = 0
\end{aligned}$$

B Proofs

B.1 Supporting Lemmas

Lemma 4 (Fan's Minimax Theory)**.** *Suppose the following conditions hold:*

1. *X be a compact Hausdorff space and Y a nonempty set (not necessarily topologized).*

2. *Let $f : X \times Y \to \mathbb{R}$ be a real-valued function.*

3. *For $\forall\ y \in Y$, $f(\cdot,y)$ is convexlike on X, i.e., for all $x_1, x_2 \in X$ and $\lambda \in [0,1]$, there exists $x_0 \in X$ such that $f(x_0,y) \leq \lambda f(x_1,y) + (1-\lambda)f(x_2,y)$.*

4. *For $\forall\ x \in X$, $f(x,\cdot)$ is concavelike on Y, i.e., for all $y_1, y_2 \in Y$ and $\lambda \in [0,1]$, there exists $y_0 \in Y$ such that $f(x,y_0) \geq \lambda f(x,y_1) + (1-\lambda)f(x,y_2)$.*

5. *For $\forall\ y \in Y$, $f(\cdot,y)$ is lower semicontinuous on X.*

Then, we have:

$$\sup_Y \inf_X f(x,y) = \inf_X \sup_Y f(x,y)$$

Proof. See Ricceri and Simons (2013) Theorem 1.3. □

Lemma 5. *Let $\{\mathcal{A}_n\}$ be a sequence of compact sets such that $d_H(\mathcal{A}_n, \mathcal{A}) = o_p(1)$ where $\mathcal{A}$ is compact, and $\mathcal{A}_n, \mathcal{A} \subset \mathcal{X}$ where $\mathcal{X}$ is a metric space. Then, the following holds:*

- *(Consistency) If $f : \mathcal{X} \to \mathbb{R}$ is continuous, then: $|\inf_{\mathcal{A}_n} f - \inf_{\mathcal{A}} f| = o_p(1)$.*
- *(Convergence Rate) If $d_H(\mathcal{A}_n, \mathcal{A}) = O_p(c_n)$ for some $c_n \to 0$, and f is Lipschitz continuous, then: $|\inf_{\mathcal{A}_n} f - \inf_{\mathcal{A}} f| = O_p(c_n)$.*

Proof. As $\mathcal{A}_n$ and $\mathcal{A}$ are compact and f is continuous, the infimum is achieved by the extreme value theorem. Denote minimizers as $a_n^* \in \mathcal{A}_n$ and $\alpha^* \in \mathcal{A}$.

1. As $d_H(\mathcal{A}_n, \mathcal{A}) = o_p(1)$, there exists a sequence $a_n \in \mathcal{A}$ such that $d(a_n^*, a_n) = o_p(1)$. By the continuity of f, we have: $|f(a_n^*) - f(a_n)| = o_p(1)$, which implies: $f(a_n^*) = f(a_n) + o_p(1) \geq f(a^*) + o_p(1)$. Similarly, there exists a sequence $b_n \in \mathcal{A}_n$ such that $d(b_n, a^*) = o_p(1)$. By the continuity of f, we have: $|f(b_n) - f(a^*)| = o_p(1)$, which implies: $f(a^*) = f(b_n) + o_p(1) \geq f(a_n^*) + o_p(1)$. Combining both inequalities, we have $f(a_n^*) \geq f(a^*) + o_p(1) \geq f(a_n^*) + o_p(1) + o_p(1)$, which implies: $|\inf_{\mathcal{A}_n} f - \inf_{\mathcal{A}} f| = |f(a_n^*) - f(a^*)| = o_p(1)$.
2. For the second part, let L be the Lipschitz constant. There exists a sequence $a_n \in \mathcal{A}$ such that $d(a_n^*, a_n) = O_p(c_n)$, by the Lipschitz continuity of f, we have: $|f(a_n^*) - f(a_n)| \leq Ld(a_n^*, a_n) = O_p(c_n)$, which implies: $f(a_n^*) \geq f(a_n) - Ld(a_n^*, a_n) \geq f(a^*) - Ld(a_n^*, a_n)$. Similarly, there exists a sequence $b_n \in \mathcal{A}_n$ such that $d(b_n, a^*) = O_p(c_n)$. By the Lipschitz continuity of f, we have: $|f(b_n) - f(a^*)| \leq Ld(b_n, a^*) = O_p(c_n)$, which implies: $f(a^*) \geq f(b_n) - Ld(b_n, a^*) \geq f(a_n^*) - Ld(b_n, a^*)$. Combining both inequalities, we have: $f(a_n^*) \geq f(a^*) - Ld(b_n, a^*) \geq f(a_n^*) - 2Ld(b_n, a^*)$, which implies: $|f(a_n^*) - f(a^*)| \leq 2Ld(b_n, a^*) = O_p(c_n)$.

□

Lemma 6. *Let $\delta > 0$ and $F_0 \in \mathcal{P}(\mathcal{U})$. Let $\mathcal{F}_{KL} := \{F \in \mathcal{P}(\mathcal{U}) \mid D_{KL}(F \| F_0) \leq \delta\}$.*

(i) *$\mathcal{F}_{KL}$ is compact in the topology of weak convergence.*

(ii) *$\mathcal{F}_{KL}$ is closed in the topology of weak convergence.*

Proof. 1. See Pinski et al. (2015) Proposition 2.1.

2. By Nutz (2021) Lemma 1.3, $D_{KL}(F\|F_0)$ is lower-semicontinuous in the topology of weak convergence, i.e., for $F_n \to F$ weakly, we have $\liminf_{n\to\infty} D_{KL}(F_n\|F_0) \geq D_{KL}(F\|F_0)$. Since $F_n \in \mathcal{F}_{KL}$, we have $D_{KL}(F\|F_0) \leq \delta$.

□

Lemma 7. *Let $\nu_i \in \mathcal{P}(\mathcal{U}_i)$ for $i \in \{1, \cdots, k\}$, then:*

(i) $\Pi(\nu_1, \cdots, \nu_k)$ *is closed in the topology of weak convergence.*

(ii) $\Pi(\nu_1, \cdots, \nu_k)$ *is compact and convex in the topology of weak convergence.*

(iii) $\Pi(\nu_1, \cdots, \nu_k)$ *is a Hausdorff topological space.*

(iv) Under Assumption 2(i), $\Pi(\nu_1, \nu_k)$ *is closed in the topology of weak convergence.*

(v) $\Pi(\nu_1, \nu_k)$ *is convex in the topology of weak convergence.*

(vi) Under Assumption 2(i), $\Pi(\nu_1, \nu_k)$ *is compact in the topology of weak convergence.*

(vii) $\Pi(\nu_1, \nu_k)$ *is a Hausdorff topological space.*

(viii) Suppose $\mathcal{N}$ is convex and closed and Assumption 2(i) holds, $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$ *is closed in the topology of weak convergence.*

(ix) Suppose $\mathcal{N}$ is convex and closed and Assumption 2(i) holds, then $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$ *is convex in the topology of weak convergence.*

(x) Suppose $\mathcal{N}$ is convex and closed and Assumption 2(i) holds, $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$ *is compact in the topology of weak convergence.*

(xi) Suppose $\mathcal{N}$ is convex and closed and Assumption 2(i) holds, $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$ *is a Hausdorff topological space.*

(xii) Lemmas 7(viii)-7(xi) also hold for $\{F \in \mathcal{P}(\Xi^2) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}\}$.

Proof. 1. **Closedness and compactness of** $\Pi(\nu_1, \cdots, \nu_k)$**:** see the proof of Villani et al. (2009) Theorem 4.1.

2. **Closedness and compactness of** $\Pi(\nu_1, \nu_k)$**:** By Assumption 2(i), $\Pi(\nu_1, \nu_k)$ is tight, and by Prokhorov's theorem it has a compact closure. By passing to the limit in the equation for marginals, $\Pi(\nu_1, \nu_k)$ is closed. Therefore, it is compact.

3. **Closedness and compactness of $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$:** By Assumption 2(i), the set $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$ is tight, and by Prokhorov's theorem it has a compact closure. By passing to the limit in the equation for marginals, it is closed. Therefore, it is compact.

4. **Closedness and compactness of $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}\}$:** The proof is similar to the previous part.

5. **Convexity:** It is straightforward.

6. **Hausdorff:** By Billingsley (2013) Page 72(i), the Prohorov distance is a metric on the space of probability measures. Metrizable topological spaces are Hausdorff.

□

Lemma 8. *Let $\delta \geq 0$, $F_0 \in \mathcal{P}(\mathcal{U})$, and $\nu_i \in \mathcal{P}(\mathcal{U}_i)$ for $i \in \{1, \cdots, k\}$. Define:*

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \cdots, \nu_k), D_{KL}(F \| F_0) \leq \delta\}$$

(i) $\mathcal{F}$ is closed and compact in the topology of weak convergence.

(ii) $\mathcal{F}$ is convex in the topology of weak convergence.

(iii) $\mathcal{F}$ is a Hausdorff topological space.

Proof. 1. By Lemmas 6(ii) and 7(i), $\mathcal{F}$ is the intersection of two closed sets. Therefore, it is closed. By Rudin et al. (1976) Theorem 2.35, $\mathcal{F}$ is compact.

2. Since KL divergence is jointly convex (see Nutz, 2021, Lemma 1.3), we have $D_{KL}(\lambda F_1 + (1-\lambda)F_2 \| F_0) \leq \lambda D_{KL}(F_1 \| F_0) + (1-\lambda) D_{KL}(F_2 \| F_0) \leq \delta$ for $\lambda \in [0,1]$ and $F_1, F_2 \in \mathcal{F}$. Moreover, $\lambda F_1 + (1-\lambda)F_2 \in \Pi(\nu_1, \cdots, \nu_k)$. Therefore, $\mathcal{F}$ is convex.

3. By Billingsley (2013) Page 72(i), the Prohorov distance is a metric on the space of probability measures. Metrizable topological spaces are Hausdorff.

□

Lemma 9. *Suppose Assumption 2(i) holds. Let $\delta \geq 0$, $F_0 \in \mathcal{P}(\mathcal{U})$, and $\nu_i \in \mathcal{P}(\mathcal{U}_i)$ for $i \in \{1, k\}$. Let $\mathcal{F}_{relaxed} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \nu_k), D_{KL}(F \| F_0) \leq \delta\}$.*

(i) $\mathcal{F}_{relaxed}$ is closed and compact in the topology of weak convergence.

(ii) $\mathcal{F}_{relaxed}$ is convex in the topology of weak convergence.

(iii) $\mathcal{F}_{relaxed}$ *is a Hausdorff topological space.*

Proof. 1. By Lemmas 6(ii) and 7(iv), $\mathcal{F}_{\text{relaxed}}$ is the intersection of two closed sets. Therefore, it is closed. By Rudin et al. (1976) Theorem 2.35, $\mathcal{F}_{\text{relaxed}}$ is compact.

2. The proof is identical to the proof of Lemma 8(ii).

3. The proof is identical to the proof of Lemma 8(iii).

□

Lemma 10. *Suppose Assumption 2(i) holds. Let $\delta \geq 0$, $F_0 \in \mathcal{P}(\mathcal{U})$ and $\nu_k \in \mathcal{P}(\mathcal{U}_k)$. Suppose $\mathcal{N} \subseteq \mathcal{P}(\mathcal{U}_1)$ is convex and closed. Let $\mathcal{F}_{\mathcal{N},Relaxed} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}, D_{KL}(F \| F_0) \leq \delta\}$.*

(i) $\mathcal{F}_{\mathcal{N},Relaxed}$ *is closed and compact in the topology of weak convergence.*

(ii) $\mathcal{F}_{\mathcal{N},Relaxed}$ *is convex in the topology of weak convergence.*

(iii) $\mathcal{F}_{\mathcal{N},Relaxed}$ *is a Hausdorff topological space.*

Proof. 1. By Lemma 7(viii), and Rudin et al. (1976) Theorem 2.35, $\mathcal{F}_{\mathcal{N},\text{Relaxed}}$ is compact.

2. Note that $\mathcal{N}$ is convex. For any $F_1, F_2 \in \mathcal{F}_{\mathcal{N},\text{Relaxed}}$, we have $\nu_1, \nu_2 \in \mathcal{N}$. By the convexity of $\mathcal{N}$, we have $\lambda\nu_1 + (1-\lambda)\nu_2 \in \mathcal{N}$ for $\lambda \in [0, 1]$. Therefore, $\lambda F_1 + (1-\lambda)F_2 \in \Pi(\lambda\nu_1 + (1-\lambda)\nu_2, \nu_k)$ and $D_{KL}(\lambda F_1 + (1-\lambda)F_2 \| F_0) \leq \delta$. Thus, $\mathcal{F}_{\mathcal{N},\text{Relaxed}}$ is convex.

3. The proof is identical to the proof of Lemma 8(iii).

□

B.2 Proofs in Section 2

B.2.1 Proof of Theorem 1

Recall the Lagrangian of the Primal is:

$$\kappa(\delta, P) = \inf_{(\theta,v,F)\in\Theta\times\mathcal{V}\times\mathcal{F}} \sup_{\lambda\in\mathbb{R}^{d_P},g\in\mathcal{G}} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] - \lambda^T P.$$

Lemma 11 (Minimax)**.** *Under Assumption 1, for every $\delta > 0$, we have:*

$$\kappa(\delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P},g\in\mathcal{G}} \inf_{F\in\mathcal{F}} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right] - \lambda^T P.$$

Proof. Fix (θ, v). We verify the conditions in Lemma 4 for

$$\mathcal{L}_0(F, g, \lambda) := \mathbb{E}_F\left[c(U; \theta, v, g, \lambda)\right] - \lambda^T P, \qquad F \in \mathcal{F}, \quad (\lambda, g) \in \mathbb{R}^{d_P} \times \mathcal{G}.$$

By Lemma 8(i), $\mathcal{F}$ is compact; by Lemma 8(ii), $\mathcal{F}$ is convex; and by Lemma 8(iii), $\mathcal{F}$ is Hausdorff. For fixed (λ, g), $\mathcal{L}_0(F, g, \lambda)$ is affine, hence convexlike, in F. For fixed F, $\mathcal{L}_0(F, g, \lambda)$ is affine, hence concavelike, in (λ, g) because $\mathcal{G}$ is convex and $\psi(U; \theta, v, g)$ is linear in g. Finally, let $h(U) := -(1 + \|\lambda\|_1) C_{\theta,v,g}(1 + d(U, \hat{U}))$. Under Assumption 1(v), $h \in L^1(F)$ for all $F \in \mathcal{F}$. Therefore, for a given (λ, g), $\mathbb{E}_F\left[c(U; \theta, v, g, \lambda)\right]$ is lower-semicontinuous in F by Villani et al. (2009) Lemma 4.3. Thus, Lemma 4 applies. □

Lemma 12 (KL-radius duality)**.** *Fix (θ, v, λ, g) and suppose $\delta > 0$. Under Assumption 1, we have:*

$$\inf_{F \in \mathcal{F}} \mathbb{E}_F\left[c(U; \theta, v, g, \lambda)\right] = \sup_{\lambda_{KL} \geq 0} \left\{ \inf_{F \in \Pi(\nu_1, \ldots, \nu_k)} \mathbb{E}_F\left[c(U; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta \right\}.$$

Proof. Since $F_0 \in \Pi(\nu_1, \ldots, \nu_k)$ by Assumption 1(ii) and $\delta > 0$, Slater's condition holds:

$$D_{KL}(F_0 \| F_0) = 0 < \delta.$$

The objective $F \mapsto \mathbb{E}_F[c(U; \theta, v, g, \lambda)]$ is affine, and $F \mapsto D_{KL}(F \| F_0)$ is convex and lower-semicontinuous. The growth condition gives finite values at the Slater point and rules out value $-\infty$. Therefore, standard scalar convex Lagrangian duality under Slater's condition (Rockafellar, 1974, Theorems 17 and 18(a)) gives no duality gap, which gives the displayed equality. □

Combining Lemma 11 and Lemma 12, we obtain:

$$\kappa(\delta, P) = \inf_{(\theta,v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, g \in \mathcal{G}, \lambda_{KL} \geq 0} \inf_{F \in \Pi(\nu_1, \ldots, \nu_k)} \mathbb{E}_F\left[c(U; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P.$$

Lemma 13. *Define:*

$$\mathcal{L}(\delta, \theta, v, g, \lambda) := \sup_{\lambda_{KL} \geq 0} \inf_{F \in \Pi(\nu_1, \ldots, \nu_k)} \mathbb{E}_F\left[c(U; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

Under Assumption 1, we have:

$$\mathcal{L}(\delta, \theta, v, g, \lambda) = \sup_{\lambda_{KL} \geq 0} \inf_{F \in \Pi(\nu_1, \cdots, \nu_k)} \mathbb{E}_F\left[c(U; \theta, v, g, \lambda) + \lambda_{KL} \rho(U)\right] + \lambda_{KL}(D_{KL}(F \| F_{\otimes}) - \delta) - \lambda^T P$$

Proof. By Assumption 1(ii), we have:

$$D_{KL}(F\|F_0) = \int \log(\frac{dF}{dF_\otimes}\frac{dF_\otimes}{dF_0})dF = \int \log(\frac{dF}{dF_\otimes})dF + \int \log(\frac{dF_\otimes}{dF_0})dF = D_{KL}(F\|F_\otimes) + \int \rho(U)dF$$

Moreover, by Assumption 1(v), we have $c(U;\theta,v,g,\lambda), \rho \in L^1(F)$ for $\forall\ F \in \Pi(\nu_1,\cdots,\nu_k)$. Therefore, the sum of two expectations is also well-defined. $\square$

Lemma 14. *Suppose Assumption 1 holds. If $c(U;\theta,v,g,\lambda)$ and $\rho(U)$ are continuous in U, then optimizing over $\lambda_{KL} > 0$ gives the same value as optimizing over $\lambda_{KL} \geq 0$, i.e.,*

$$\mathcal{L}(\delta,\theta,v,g,\lambda) = \sup_{\lambda_{KL}>0} \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda) + \lambda_{KL}\rho(U)\right] + \lambda_{KL}(D_{KL}(F\|F_\otimes) - \delta) - \lambda^T P$$

Proof. By Eckstein and Nutz (2023) Proposition 3.1, we have:

$$\lim_{\lambda_{KL}\downarrow 0} \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda) + \lambda_{KL}\rho(U)\right] + \lambda_{KL}(D_{KL}(F\|F_\otimes) - \delta) = \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right]$$

Therefore, for any sequence $\lambda_{KL}^i \to 0$ as $i \to \infty$, we have:

$$\lim_{i\to\infty} \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda) + \lambda_{KL}\rho(U)\right] + \lambda_{KL}^i(D_{KL}(F\|F_\otimes) - \delta) = \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda)\right]$$

$\square$

Then, we show the EOT duality part. For $\lambda_{KL} > 0$, consider the following EOT problem:

$$\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) := \inf_{F\in\Pi(\nu_1,\cdots,\nu_k)} \mathbb{E}_F\left[c(U;\theta,v,g,\lambda) + \lambda_{KL}\rho(U)\right] + \lambda_{KL}D_{KL}(F\|F_\otimes)$$

Theorem 10 (EOT Duality)**.** *Under Assumption 1, for $\lambda_{KL} > 0$, the following holds:*

$$\mathcal{C}(\theta,v,g,\lambda,\lambda_{KL}) = \sup_{\{\phi_i\in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i}\phi_i(U_i) - \lambda_{KL}\mathbb{E}_{F_0}\exp(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}}) + \lambda_{KL}$$

Moreover, the worst-case distribution is given by:

$$\frac{dF^*(U)}{dF_0(U)} = \exp(\frac{\sum_{i=1}^k \phi_i^*(U_i) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}}) \quad F_0\text{-}a.s.$$

where $\{\phi_i^*\}_{i=1}^k$ are unique maximizers up to additive constants F_0-almost surely.

Proof. Under Assumption 1(v), $c(U;\theta,v,g,\lambda), \rho(U) \in L^1(F_\otimes)$. For two marginals ($k = 2$), see Nutz (2021) Theorem 4.7 and Remark 4.8(a). The results generalize directly to the multi-marginal case (see Nutz and Wiesel, 2022, Section 6). Their results hold $F_\otimes$-a.s. Since $F_0 \ll F_\otimes$, the results also hold F_0-a.s. Moreover, note that:

$$\mathbb{E}_{F_\otimes} \exp(\frac{\sum_{i=1}^{k} \phi_i(U_i) - c(U;\theta,v,g,\lambda) - \lambda_{KL}\rho(U)}{\lambda_{KL}}) = \mathbb{E}_{F_0} \exp(\frac{\sum_{i=1}^{k} \phi_i(U_i) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}})$$

□

B.2.2 Proof of Proposition 1

Proof of Proposition 1. Write $\tilde{c}(U) := c(U;\theta,v,g,\lambda)+\lambda_{KL}\rho(U)$. Since $\rho(U) = \log \frac{dF_\otimes(U)}{dF_0(U)}$, we have

$$D_{KL}(F\|F_0) = D_{KL}(F\|F_\otimes) + \int \rho(U)dF(U).$$

Hence the EOT problem can be written relative to the product reference $F_\otimes$ with effective cost $\tilde{c}$. As shown in Nutz (2021) equation 4.11, the fact that F^* in Theorem 10 is a probability measure with marginals $(\nu_1, \cdots, \nu_k)$ implies the Schrödinger equations: for each $j \leq k$,

$$\phi_j^*(U_j) = -\lambda_{KL} \log \mathbb{E}_{F_{\otimes,-j}} \exp\left(\frac{\sum_{i\neq j} \phi_i^*(U_i) - \tilde{c}(U)}{\lambda_{KL}}\right) \quad \nu_j\text{-a.s.}$$

By the envelope condition in Proposition 1, the Leibniz integral rule applies up to order q to the Schrödinger integral. Therefore, the integral inside the logarithm is q-times continuously differentiable in U_j. The integral is strictly positive, so its logarithm is also q-times continuously differentiable. Thus ϕ_j^* admits a C^q version for each $j \leq k$. Finally,

$$\frac{dF^*(U)}{dF_0(U)} = \exp\left(\frac{\sum_{i=1}^{k} \phi_i^*(U_i) - c(U;\theta,v,g,\lambda)}{\lambda_{KL}}\right),$$

and the right-hand side is q-times continuously differentiable in U because the potentials and $c(U;\theta,v,g,\lambda)$ are q-times continuously differentiable. □

B.2.3 Proof of Theorem 2

Proof of Theorem 2. Define the set:

$$\bar{\mathcal{F}} := \left\{F \in \mathcal{P}(\Xi^2) : F \in \Pi(\nu,\nu) \text{ for some } \nu \in \mathcal{N}\right\}.$$

Then $\mathcal{F} = \left\{F \in \bar{\mathcal{F}} : D_{KL}(F \| F_0) \le \delta\right\}$. By Lemma 6 and lemma 7(xii), $\mathcal{F}$ is compact and Hausdorff; it is also convex because $\bar{\mathcal{F}}$ is convex by Lemma 7(xii) and KL divergence is convex. Fix (θ, v) and define

$$\mathcal{L}_0(F, g, \lambda) := \mathbb{E}_F \left[c(\xi, \xi'; \theta, v, g, \lambda)\right] - \lambda^T P.$$

For fixed (λ, g), $\mathcal{L}_0(F, g, \lambda)$ is affine, hence convexlike, in F. For fixed F, $\mathcal{L}_0(F, g, \lambda)$ is affine, hence concavelike, in (λ, g) because $\mathcal{G}$ is convex and $\psi(\xi, \xi'; \theta, v, g)$ is linear in g. Moreover, by Assumption 1(v) and compactness of Ξ, the minorant $h(\xi, \xi') := -(1 + \|\lambda\|_1) C_{\theta,v,g}(1 + d(\xi, \xi', \hat{\xi}, \hat{\xi}'))$ is bounded. Therefore, for given (λ, g), $F \mapsto \mathbb{E}_F \left[c(\xi, \xi'; \theta, v, g, \lambda)\right]$ is lower-semicontinuous by Villani et al. (2009) Lemma 4.3. Hence Lemma 4 gives

$$\kappa_{stationary}(\delta_1, \delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P}, g\in\mathcal{G}} \inf_{F\in\mathcal{F}} \mathbb{E}_F \left[c(\xi, \xi'; \theta, v, g, \lambda)\right] - \lambda^T P.$$

Next fix (θ, v, λ, g). Since $F_0 \in \Pi(\nu_0, \nu_0)$ and $\nu_0 \in \mathcal{N}$, we have $F_0 \in \bar{\mathcal{F}}$. Since $\delta > 0$,

$$D_{KL}(F_0 \| F_0) = 0 < \delta,$$

so Slater's condition holds for the KL constraint over $\bar{\mathcal{F}}$. The growth condition gives finite values at the Slater point and rules out value $-\infty$. Thus scalar convex Lagrangian duality under Slater's condition (Rockafellar, 1974, Theorems 17(a) and 18(a)) yields

$$\inf_{F\in\mathcal{F}} \mathbb{E}_F \left[c(\xi, \xi'; \theta, v, g, \lambda)\right] = \sup_{\lambda_{KL}\ge 0} \left\{ \inf_{F\in\bar{\mathcal{F}}} \left(\mathbb{E}_F \left[c(\xi, \xi'; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0)\right) - \lambda_{KL}\delta \right\}.$$

Since $\bar{\mathcal{F}}$ is the union of $\Pi(\nu, \nu)$ over $\nu \in \mathcal{N}$, we have

$$\inf_{F\in\bar{\mathcal{F}}} \left(\mathbb{E}_F \left[c(\xi, \xi'; \theta, v, g, \lambda)\right] + \lambda_{KL} D_{KL}(F \| F_0)\right) = \inf_{\nu\in\mathcal{N}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu).$$

Combining the preceding displays gives

$$\kappa_{stationary}(\delta_1, \delta, P) = \inf_{(\theta,v)\in\Theta\times\mathcal{V}} \sup_{\lambda\in\mathbb{R}^{d_P}, \lambda_{KL}\ge 0, g\in\mathcal{G}} \inf_{\nu\in\mathcal{N}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) - \lambda_{KL}\delta - \lambda^T P.$$

For $\lambda_{KL} > 0$, the EOT duality in Theorem 10, applied pointwise at a fixed ν, gives

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) = \sup_{\phi_1,\phi_2\in L^\infty(\Xi)} \mathbb{E}_\nu \left[\phi_1(\xi) + \phi_2(\xi)\right] - \lambda_{KL} \mathbb{E}_{F_0} \exp\left(\frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}}\right) + \lambda_{KL}.$$

The replacement of $L^1(\nu)$ by the common space $L^\infty(\Xi)$ is without loss here. Indeed, compactness of Ξ and Assumption 1(v) imply that the effective EOT cost is bounded for fixed $(\theta, v, g, \lambda, \lambda_{KL}, \nu)$. Hence the EOT potentials can be chosen bounded after normalization by (Nutz, 2021, Lemma 4.9). Moreover, using the Schrödinger equations and the reverse Fatou lemma, these bounded potentials can be chosen lower semicontinuous. Let

$$\mathcal{H} := \{\phi : \Xi \to \mathbb{R} : \phi \text{ is bounded and lower semicontinuous}\}.$$

We now apply Lemma 4 to interchange $\inf_{\nu\in\mathcal{N}}$ and $\sup_{\phi_1,\phi_2\in\mathcal{H}}$. The set $\mathcal{N}$ is compact by Lemma 6 and convex by convexity of KL divergence. For fixed $(\phi_1, \phi_2) \in \mathcal{H}^2$, the map $\nu \mapsto \mathbb{E}_\nu[\phi_1(\xi) + \phi_2(\xi)]$ is lower-semicontinuous by Villani et al. (2009) Lemma 4.3, and the remaining exponential term is independent of ν. For fixed ν, the objective is concave in (ϕ_1, ϕ_2) because it is the sum of a linear term and the negative of a convex exponential term. Thus,

$$\inf_{\nu\in\mathcal{N}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) = \sup_{\phi_1,\phi_2\in\mathcal{H}} \inf_{\nu\in\mathcal{N}} \left\{\mathbb{E}_\nu\left[\phi_1(\xi) + \phi_2(\xi)\right] - \lambda_{KL}\mathbb{E}_{F_0} \exp\left(\frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}}\right)\right.$$

Finally, for $h := \phi_1 + \phi_2$, KL-divergence DRO duality (Hu and Hong, 2013, Theorem 1) gives

$$\inf_{\nu\in\mathcal{N}} \mathbb{E}_\nu[h(\xi)] = \sup_{\eta\geq 0} \left\{-\eta \log \mathbb{E}_{\nu_0} \exp\left(-\frac{h(\xi)}{\eta}\right) - \eta\delta_1\right\},$$

□

B.3 Proofs in Section 3

We only prove Theorem 3.

B.3.1 Proof of Theorem 3

Proof of Theorem 3. 1. **Minimax Part:** The minimax part follows the minimax and KL-radius duality steps in the proof of Theorem 1. Replace $\Pi(\nu_1, \ldots, \nu_k)$ by $\Pi(\nu_1, \nu_T)$, $\mathcal{F}$ by $\mathcal{F}_{\text{relaxed}}$, and Lemmas 8(i) to 8(iii) by Lemmas 7(v) to 7(vii). The growth condition used there is replaced here by the boundedness condition in Assumption 2(iii), and Slater's condition holds because $F_0 \in \Pi(\nu_1, \nu_T)$ and $\delta > 0$.

2. **Duality Part:** For notational simplicity, let $c(U) := c(U; \theta, v, g, \lambda)$. Then, S_{dyn} can

be rewritten as:

$$\inf_{F\in\Pi(\nu_1,\nu_T)} \mathbb{E}_F\left[c(U)\right] + \lambda_{KL} D_{KL}(F\|F_0) \tag{18}$$

By Assumption 2(iii), $\exp(\frac{-c(U)}{\lambda_{KL}})$ is bounded and in particular, $\exp(\frac{-c(U)}{\lambda_{KL}}) \in L^1(F_0)$. Therefore, we can define the auxiliary reference measure R as follows:

$$dR(U) := \exp(\frac{-c(U)}{\lambda_{KL}})dF_0(U)$$

Note that $R \sim F_0$. Then, (18) is equivalent to:[34]

$$\inf_{F\in\Pi(\nu_1,\nu_T)} \lambda_{KL} D_{KL}(F\|R) \tag{19}$$

By Léonard (2014) Theorem 2.4, we have:

$$D_{KL}(F\|R) = D_{KL}(F_{1,T}\|R_{1,T}) + \mathbb{E}_{F_{1,T}}\left[D_{KL}(F_{|1,T}\|R_{|1,T})\right]$$

where $F_{1,T}$ and $R_{1,T}$ are the two-period marginals of F and R, respectively. $F_{|1,T}, R_{|1,T}$ are the conditional distribution given (ξ_1, ξ_T). In particular, we have:

$$dR_{1,T}(\xi_1,\xi_T) := \int_{\xi_2\cdots\xi_{T-1}} dR(\xi_1,\cdots,\xi_T) = \int_{\xi_2,\ldots,\xi_{T-1}} \exp\left(\frac{-c(\xi_1,\ldots,\xi_T)}{\lambda_{KL}}\right) dF_0(\xi_1,\ldots,\xi_T)$$

The second term is minimized at $F^*_{|1,T} = R_{|1,T}$. Therefore, (19) can be reduced to the static Schrödinger bridge problem:

$$\inf_{F_{1,T}\in\tilde{\Pi}(\nu_1,\nu_T)} D_{KL}(F_{1,T}\|R_{1,T}) \tag{20}$$

where $\tilde{\Pi}(\nu_1,\nu_T)$ is the set of all joint distributions of (ξ_1,ξ_T) with marginals ν_1 and ν_T. Note that:

$$dR_{1,T}(\xi_1,\xi_T) = \left(\int_{\xi_2,\ldots,\xi_{T-1}} \exp\left(\frac{-c(\xi_1,\ldots,\xi_T)}{\lambda_{KL}}\right) dF_0(\xi_2,\ldots,\xi_{T-1}|\xi_1,\xi_T)\right) dF_0^{1,T}(\xi_1,\xi_T)$$

By Assumption 2(iii), we have $R_{1,T} \sim F_0^{1,T}$ and the density $\frac{dR_{1,T}}{dF_0^{1,T}}$ is bounded above and

[34] This is the Schrödinger bridge problem, see Léonard (2013) for continuous time setting and De Bortoli et al. (2021) for discrete time setting.

bounded away from zero. By Assumption 2(ii), $D_{KL}(\nu_1 \otimes \nu_T \| F_0^{1,T}) < +\infty$. Therefore,

$$D_{KL}(\nu_1 \otimes \nu_T \| R_{1,T}) \leq D_{KL}(\nu_1 \otimes \nu_T \| F_0^{1,T}) + C < +\infty$$

for some finite constant C. Hence $\nu_1 \otimes \nu_T \in \Pi_{\text{fin}}(\nu_1, \nu_T) := \{F \in \tilde{\Pi}(\nu_1, \nu_T) \mid D_{KL}(F \| R_{1,T}) < +\infty\}$, so $\Pi_{\text{fin}}(\nu_1, \nu_T) \neq \emptyset$. By Assumption 2(ii) and Nutz (2021) Theorem 2.1, the unique solution to (20) has the form:

$$\frac{dF^*_{1,T}(\xi_1, \xi_T)}{dR_{1,T}(\xi_1, \xi_T)} = \exp\left(\phi_1^*(\xi_1) + \phi_T^*(\xi_T)\right) \quad R_{1,T}\text{-a.s.}$$

and $\phi_1^* \in L^1(\nu_1)$, $\phi_T^* \in L^1(\nu_T)$. By Nutz (2021) Theorem 3.2, we have:

$$\inf_{F_{1,T} \in \tilde{\Pi}(\nu_1, \nu_T)} D_{KL}(F_{1,T} \| R_{1,T}) = \sup_{\phi_1 \in L^1(\nu_1), \phi_T \in L^1(\nu_T)} \mathbb{E}_{\nu_1}\phi_1 + \mathbb{E}_{\nu_T}\phi_T - \int \exp\left(\phi_1 + \phi_T\right) dR_{1,T} + 1$$

Multiplying both sides by λ_{KL} and letting $\phi_1 := \lambda_{KL}\phi_1$ and $\phi_T := \lambda_{KL}\phi_T$, we have:

$$\inf_{F_{1,T} \in \tilde{\Pi}(\nu_1, \nu_T)} \lambda_{KL} D_{KL}(F_{1,T} \| R_{1,T}) = \sup_{\phi_1 \in L^1(\nu_1), \phi_T \in L^1(\nu_T)} \mathbb{E}_{\nu_1}\phi_1 + \mathbb{E}_{\nu_T}\phi_T - \lambda_{KL}\mathbb{E}_{R_{1,T}} \exp\left(\frac{\phi_1 + \phi_T}{\lambda_{KL}}\right) + \lambda_{KL}$$

Then, we have:

$$dF^*(U) = dF^*_{1,T} dF^*_{|1,T} = \exp\left(\frac{\phi_1^*(\xi_1) + \phi_T^*(\xi_T) - c(U)}{\lambda_{KL}}\right) dF_0(U)$$

where ϕ_1^* and ϕ_T^* are the unique maximizers up to an additive constant.

3. **Markov Property:** Assumption 2(iv) implies that $c(U)$ is also pairwise additive, i.e., $c(U) = \sum_{t=1}^{T-1} c_t(\xi_t, \xi_{t+1})$ for some $c_t(\xi_t, \xi_{t+1})$. Therefore, we can write:

$$dF^*(U) = \exp\left(\sum_{t=1}^{T-1} \frac{-c_t(\xi_t, \xi_{t+1})}{\lambda_{KL}} + \frac{\phi_1^*(\xi_1) + \phi_T^*(\xi_T)}{\lambda_{KL}}\right) dF_0(U)$$

which has Markov property as F_0 has Markov property.[35]

4. **Equivalence:** Since $\mathcal{F}_{\text{Markov}} \subseteq \mathcal{F}_{\text{relaxed}}$, we have $\kappa_{\text{TI}}(\delta, P) \geq \tilde{\kappa}_{\text{TI}}(\delta, P)$. For the reverse inequality, take any feasible $F \in \mathcal{F}_{\text{relaxed}}$ and let F^M be the Markov law with the same initial distribution and transition kernels given by the regular conditional distributions

[35]The left part is the (unnormalized) pairwise Markov random field Wainwright et al. (2008).

$F(d\xi_{t+1} \mid \xi_t)$ induced by the two-period marginals of F:

$$dF^M(U) := \nu_1(d\xi_1) \prod_{t=1}^{T-1} F(d\xi_{t+1} \mid \xi_t).$$

By construction, $F^M \in \Pi_{\text{Markov}}(\nu_1, \nu_T)$ and $F^M_{t,t+1} = F_{t,t+1}$ for each $t \leq T-1$. Therefore, by Assumption 2(iv), $\mathbb{E}_{F^M}[s(U;\theta,v)] = \mathbb{E}_F[s(U;\theta,v)], \mathbb{E}_{F^M}[m(U;\theta,v)] = \mathbb{E}_F[m(U;\theta,v)], \mathbb{E}_{F^M}[\psi(U;\theta,v,g)] = \mathbb{E}_F[\psi(U;\theta,v,g)]$ for every $g \in \mathcal{G}$. Moreover, since F_0 is first-order Markov, the chain rule for relative entropy implies

$$D_{KL}(F^M \| F_0) \leq D_{KL}(F \| F_0).$$

Hence every feasible law for the relaxed problem has a Markov feasible counterpart with the same objective value. Therefore, $\kappa_{\text{TI}}(\delta, P) \leq \tilde{\kappa}_{\text{TI}}(\delta, P)$, and the two inequalities give $\kappa_{\text{TI}}(\delta, P) = \tilde{\kappa}_{\text{TI}}(\delta, P)$.

□

B.3.2 Proof of Theorem 4

Proof of Theorem 4. The proof of minimax part is similar to Lemma 11. Lemma 7(ii) is replaced by Lemmas 7(x) and 7(ix). Lemma 7(iii) is replaced by Lemma 7(xi). Finally, the bound on the cost function is replaced by the finite constant in Assumption 2(iii). □

B.3.3 Proof of Lemma 1

Proof of Lemma 1. The EOT problem is equivalent to the Schrödinger bridge problem $\text{EOT}(\nu, \nu_T) = \inf_{F_{1,T} \in \tilde{\Pi}(\nu, \nu_T)} D_{KL}(F_{1,T} \| R_{1,T})$, where $R_{1,T}$ is fixed and does not depend on ν. Let π_1, π_2 be the unique solutions with respect to ν_1, ν_2, respectively. Then, $\pi := \lambda\pi_1 + (1-\lambda)\pi_2 \in \tilde{\Pi}(\lambda\nu_1 + (1-\lambda)\nu_2, \nu_T)$ for all $\lambda \in [0,1]$. Since π is feasible but not necessarily optimal:

$$\text{EOT}(\lambda\nu_1 + (1-\lambda)\nu_2, \nu_T) \leq D_{KL}(\pi \| R_{1,T})$$

By the joint convexity of the KL divergence (see Nutz, 2021, Lemma 1.3):

$$D_{KL}(\lambda\pi_1 + (1-\lambda)\pi_2 \| R_{1,T}) \leq \lambda D_{KL}(\pi_1 \| R_{1,T}) + (1-\lambda) D_{KL}(\pi_2 \| R_{1,T})$$

Combining the two displays:

$$\mathrm{EOT}(\lambda\nu_1 + (1-\lambda)\nu_2, \nu_T) \le \lambda\,\mathrm{EOT}(\nu_1, \nu_T) + (1-\lambda)\,\mathrm{EOT}(\nu_2, \nu_T)$$

See Goldfeld et al. (2024) Lemma E.23 for the directional derivative. □

B.4 Proofs in Section 4

B.4.1 Proof of Lemma 2

Proof of Lemma 2. By the continuity of $\mathbb{E}_F[m(U;\theta, v(\alpha))]$, $\mathcal{A}_I, \hat{\mathcal{A}}_I$ are closed. By Rudin et al. (1976) Theorem 2.35, a closed subset of a compact set is compact. Therefore, $\mathcal{A}_I, \hat{\mathcal{A}}_I$ are compact as $\mathcal{A}$ is compact (Lemmas 8(i) and 9(i)). For the nonempty part, see the proof of Theorem 5. □

B.4.2 Proof of Theorem 5

Proof of Theorem 5(i). 1. Since $\epsilon_n \ge \|P_0 - P_n\|_\infty$, we have $\mathcal{A}_I \subseteq \hat{\mathcal{A}}_I$.

2. If $\mathcal{A}_I = \mathcal{A}$, the result is trivial.

3. Suppose $\mathcal{A}_I \ne \mathcal{A}$. By Assumption 3(v), we have for some $\delta(\varepsilon) > 0$:

$$\inf_{d(\alpha,\mathcal{A}_I)>\varepsilon} \|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_n\|_\infty \ge \inf_{d(\alpha,\mathcal{A}_I)>\varepsilon} \|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_0 + o_p(1)\|_\infty \ge \delta(\varepsilon) + o_p(1)$$

Similarly, we have:

$$\sup_{\hat{\mathcal{A}}_I} \|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_0\|_\infty \le \sup_{\hat{\mathcal{A}}_I} \|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_n + o_p(1)\|_\infty \le \frac{c_n}{\sqrt{n}} + o_p(1) = o_p(1)$$

Therefore, with probability approaching 1:

$$\sup_{\hat{\mathcal{A}}_I} \|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_0\|_\infty < \delta(\varepsilon) \le \inf_{d(\alpha,\mathcal{A}_I)>\varepsilon} \|\mathbb{E}_F[m(U;\theta,v(\alpha))] - P_n\|_\infty$$

which implies that: with probability approaching 1, $\hat{\mathcal{A}}_I \bigcap \{\alpha : d(\alpha, \mathcal{A}_I) > \varepsilon\} = \emptyset$. Thus, $\hat{\mathcal{A}}_I \subseteq \{\alpha : d(\alpha,\mathcal{A}_I) \le \varepsilon\}$ with probability approaching 1. Therefore, we have with probability approaching 1, $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) \le \varepsilon$. As ε is arbitrary, $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = o_p(1)$.

□

Lemma 15 (Existence of a Polynomial Minorant). *Under Assumptions 3, 4, and 5, we have for* $\forall\, \varepsilon \in (0,1)$ *there exists* $(\kappa_\varepsilon, n_\varepsilon)$ *such that for all* $n \geq n_\varepsilon$*, we have:*

$$\|\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_n\|_\infty \geq \frac{C_1}{2}\min\{C_2, d(\alpha,\mathcal{A}_I)\}$$

uniformly on $\{\alpha | d(\alpha,\mathcal{A}_I) \geq \frac{\kappa_\varepsilon}{\sqrt{n}}\}$ *with probability at least* $1-\varepsilon$.

Proof. Note that:

$$\begin{aligned}\sqrt{n}\|\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_n\|_\infty &= \|\sqrt{n}(\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_0) + \sqrt{n}(P_0 - P_n)\|_\infty \\ &\geq \left|\|\sqrt{n}(\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_0)\|_\infty - \|\sqrt{n}(P_0 - P_n)\|_\infty\right| \\ &= \left|\|\sqrt{n}(\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_0)\|_\infty - O_p(1)\right|\end{aligned}$$

where we used $\|x+y\| \geq |\|x\|-\|y\||$ and $\sqrt{n}(P_0-P_n) = O_p(1)$. Therefore, for $\forall\, \varepsilon$ there exists $M_\varepsilon > 0$ and $n_{\varepsilon,1}$ such that for all $n \geq n_{\varepsilon,1}$, with probability at least $1-\varepsilon$: $|O_p(1)| \leq M_\varepsilon$. Choose $(\kappa_\varepsilon, n_\varepsilon)$ such that $n_\varepsilon \geq n_{\varepsilon,1}$, $C_1\kappa_\varepsilon \geq 2M_\varepsilon$, and $\sqrt{n}C_1C_2 \geq 2M_\varepsilon$ for all $n \geq n_\varepsilon$. Then, with probability at least $1-\varepsilon$: uniformly on $\{\alpha : d(\alpha,\mathcal{A}_I) \geq \frac{\kappa_\varepsilon}{\sqrt{n}}\}$:

$$\begin{aligned}\|\sqrt{n}(\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_0)\|_\infty - O_p(1) &\geq \|\sqrt{n}(\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_0)\|_\infty - M_\varepsilon \\ &\geq \sqrt{n}C_1\min\{C_2, d(\alpha,\mathcal{A}_I)\} - M_\varepsilon \\ &\geq \frac{1}{2}\sqrt{n}C_1\min\{C_2, d(\alpha,\mathcal{A}_I)\} + \frac{1}{2}\sqrt{n}C_1\min\{C_2, \frac{\kappa_\varepsilon}{\sqrt{n}}\} - M_\varepsilon \\ &\geq \frac{1}{2}\sqrt{n}C_1\min\{C_2, d(\alpha,\mathcal{A}_I)\}\end{aligned}$$

Therefore, with probability at least $1-\varepsilon$: uniformly on $\{\alpha : d(\alpha,\mathcal{A}_I) \geq \frac{\kappa_\varepsilon}{\sqrt{n}}\}$:

$$\|\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_n\|_\infty \geq \frac{1}{2}C_1\min\{C_2, d(\alpha,\mathcal{A}_I)\}$$

□

Proof of Theorem 5(ii). For $\forall\, \varepsilon > 0$, let the positive constants $(\kappa_\varepsilon, n_\varepsilon)$ be as specified in Lemma 15. Let $\bar{c} := \max\{\frac{C_1}{2}\kappa_\varepsilon, c_n\}$. Furthermore, there exists $n_{\varepsilon'} \geq n_\varepsilon$ such that for all $n \geq n_{\varepsilon'}$, with probability at least $1-\varepsilon$: $\varepsilon_n := \frac{\bar{c}}{\frac{C_1}{2}\sqrt{n}} \leq C_2$ as $\frac{c_n}{\sqrt{n}} = o_p(1)$, and $\varepsilon_n \geq \frac{\kappa_\varepsilon}{\sqrt{n}}$. Therefore, with probability at least $1-\varepsilon$, for all $n \geq n_{\varepsilon'}$:

$$\inf_{\alpha, d(\alpha,\mathcal{A}_I)\geq\varepsilon_n} \|\mathbb{E}_F\left[m(U;\theta,v(\alpha))\right] - P_n\|_\infty \geq \frac{C_1}{2}\min\{C_2, d(\alpha,\mathcal{A}_I)\} \geq \frac{C_1}{2}\min\{C_2,\varepsilon_n\} \geq \frac{C_1}{2}\varepsilon_n = \frac{\bar{c}}{\sqrt{n}} \geq \frac{c_n}{\sqrt{n}}$$

Therefore, combining the first part in Theorem 5(i) we have: with probability at least $1-\varepsilon$: $\mathcal{A}_I \subseteq \hat{\mathcal{A}}_I \subseteq \{\alpha | d(\alpha, \mathcal{A}_I) \leq \varepsilon_n\}$. Thus, $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) \leq \varepsilon_n$. Therefore, we have: for $\forall\ \varepsilon > 0$, there exists $n_{\varepsilon'}$ such that for all $n \geq n_{\varepsilon'}$, we have: with probability at least $1-\varepsilon$: $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) \leq \varepsilon_n$. As ε is arbitrary, we have $|\frac{d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I)}{\frac{\max\{1, c_n\}}{\sqrt{n}}}| \leq \frac{\max\{\frac{C_1}{2}\kappa_\varepsilon, c_n\}}{\max\{1, c_n\}\frac{C_1}{2}} := M_{1,\varepsilon}$ with probability at least $1-\varepsilon$. Therefore, we have $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = O_p(\frac{\max\{1, c_n\}}{\sqrt{n}})$. □

B.4.3 Proof of Theorem 6

Proof. It is a direct consequence of Lemma 5 and Theorem 5 □

B.4.4 Proof of Theorem 7

Proof of Theorem 7. We verify conditions in Bonnans and Shapiro (2013) Theorem 4.25.

We first verify the setting in Bonnans and Shapiro (2013) Page 260. By Bogachev and Ruas (2007) Theorem 4.6.1, all real countably additive measures on $(\mathcal{U}, \mathcal{B}(\mathcal{U}))$ with the variation norm forms a Banach space. Therefore, the product of $\mathbb{R}^{d_\theta}$ and all real countably additive measures with the variation norm plus the Euclidean norm also forms a Banach space. Therefore, the setting is satisfied.

Moreover, by Lemmas 8(i) and 9(i), the sets $\mathcal{F}$ and $\mathcal{F}_{\text{relaxed}}$ are closed in the weak topology. Since convergence in $\|\cdot\|_{\mathbb{B}}$ implies weak convergence of the measure component, $\mathcal{F}$ and $\mathcal{F}_{\text{relaxed}}$ are also closed in the norm topology. By Lemmas 8(ii) and 9(ii), the sets $\mathcal{F}$ and $\mathcal{F}_{\text{relaxed}}$ are convex. By Assumption 3(i), Θ is also closed and convex. Therefore, $\mathcal{A}$ is closed and convex in the Banach space. According to Bonnans and Shapiro (2013) equation (2.194), the Robinson's constraint qualification (defined in their equation (2.163)) is equivalent to Assumption 8(i). In their notation, $G(\alpha, P) := P(\alpha) - P$ and $f(\alpha, P) = s(\alpha)$. Thus, the Robinson's constraint qualification holds.

By Bonnans and Shapiro (2013) Theorem 4.9. The Robinson's constraint qualification implies the directional regularity condition for any direction.

Therefore, by Bonnans and Shapiro (2013) Theorem 4.25, the maps $\kappa(\delta, P)$ and $\tilde{\kappa}_{\text{TI}}(\delta, P)$ are Hadamard directionally differentiable at P_0, and note that $D_P\mathcal{L}(\alpha, \lambda, P_0)h = -\lambda^T h$ where $\mathcal{L}(\alpha, \lambda, P) := s(\alpha) + \lambda^T(P(\alpha) - P)$.

The asymptotic distribution follows Fang and Santos (2019) Theorem 2.1. □

B.5 Proofs in Section 5

B.5.1 Proof of Theorem 8

Proof of Theorem 8. The quantization rate is in Eckstein and Nutz (2024) Remark 2.1. Other parts follow directly from Eckstein and Nutz (2024) Theorem 3.1(i). □

B.5.2 Proof of Theorem 9

Lemma 16. *Under Assumptions 3 and 9(ii), $\mathcal{A}^{\delta}_{I,Lip}$ is compact.*

Proof. By Assumption 9(ii), Π_{TH} and Π_{TI} are both tight, and by Prokhorov's theorem, they have compact closure. Moreover, the restrictions $C_3 \leq dF \leq C_4$ and $\|dF\|_{Lip} \leq L$ are closed under weak convergence on the compact space $\mathcal{U}$: indeed, by Arzelà–Ascoli, any weakly convergent sequence with uniformly bounded and uniformly Lipschitz densities admits a uniformly convergent density subsequence, and the uniform limit preserves the same bounds and Lipschitz constant. By passing to the limit in the equation for marginals, Π_{TH} and Π_{TI} are closed, which implies $\mathcal{A}^{\delta}_{Lip} = \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi, D_{KL}(F\|F_0) \leq \delta\}$ is closed where Π is either Π_{TH} or Π_{TI}. By Lemma 6(i) and Rudin et al. (1976) Theorem 2.35, $\mathcal{A}^{\delta}_{Lip}$ is compact. By Assumption 3(v), $\mathcal{A}^{\delta}_{I,Lip}$ is closed. Therefore, it is compact. □

Lemma 17. *Let Π_{Lip} be as in Assumption 9. Equip the density component with the L^∞ norm. Under Assumption 9, $D_{KL}(F\|F_0)$ is continuously differentiable on an L^∞-open neighborhood of Π_{Lip}.*

Proof. Let μ be the reference measure with respect to which the densities in Π_{Lip} are defined. Write $f = dF/d\mu$ and $f_0 = dF_0/d\mu$, and define: $K(f) := \int f \log \frac{f}{f_0} d\mu$. Since $\mathcal{U}$ is compact, $\mu(\mathcal{U}) < +\infty$. By Assumption 9, each $F \in \Pi_{Lip}$ has density f satisfying $C_3 \leq f \leq C_4$, and F_0 has density f_0 satisfying $C_3 \leq f_0 \leq C_4$. Consider the L^∞-open set: $\mathcal{O} := \{f \in L^\infty(\mu) : \operatorname{ess\,inf}_{U\in\mathcal{U}} f(U) > C_3/2\}$. For $f \in \mathcal{O}$ and $h \in L^\infty(\mu)$, the derivative of K is: $DK(f)[h] = \int \left(1 + \log \frac{f}{f_0}\right) h \, d\mu$. Indeed, for each U, the scalar function $x \mapsto x \log(x/f_0(U))$ has second derivative $1/x$, which is uniformly bounded on $x \geq C_3/2$. Hence the Taylor remainder is $O(\|h\|^2_\infty)$ uniformly on $\mathcal{U}$.

Moreover, for $F_1, F_2 \in \Pi_{Lip}$ with densities f_1, f_2 and for $h \in L^\infty(\mu)$, the mean value theorem gives: $|(DK(f_1) - DK(f_2))[h]| = \left|\int \log \frac{f_1}{f_2} h \, d\mu\right| \leq \frac{1}{C_3} \int |f_1 - f_2| \, |h| \, d\mu$. Taking the supremum over $\|h\|_\infty \leq 1$ yields:

$$\|DK(f_1) - DK(f_2)\|_{op} \leq \frac{\mu(\mathcal{U})}{C_3} \|f_1 - f_2\|_\infty.$$

Therefore, K has a Lipschitz-continuous derivative in the L^∞ density norm.

Finally, for admissible directions of the form $h = f_1 - f$, where F_1, F are probability measures, we have $\int h\, d\mu = 0$. Thus:

$$DK(f)[f_1 - f] = \int \left(1 + \log \frac{f}{f_0}\right)(f_1 - f)d\mu = -D_{KL}(F\|F_0) + \int \log \frac{dF}{dF_0} dF_1,$$

which is the directional derivative formula.[36] Restricting the L^∞ derivative to probability densities in Π_{Lip} gives the desired continuous differentiability of $F \mapsto D_{KL}(F\|F_0)$. □

Proof of Theorem 9. Consider the time-homogeneous optimization problem and set $\Pi_{Lip} := \Pi_{\mathrm{TH}}$:

$$\kappa^{Lip}(\delta, P_0) = \inf_{\alpha \in \Theta \times \Pi_{Lip}} s(\alpha) \quad \text{s.t.} \quad P(\alpha) = P_0, \quad D_{KL}(F\|F_0) \leq \delta$$

Note that Π_{Lip} is convex and closed, $\mathcal{A}^{\delta}_{I,Lip}$ is compact by Lemma 16, and $s(\alpha)$ is continuous. By the extreme value theorem, the infimum is achieved. Therefore, $\mathcal{A}^{\delta,*}_{I,Lip}$ is nonempty.

We aim to show the Hadamard directionally differentiability of $\kappa^{Lip}(\delta, P_0)$ in the directional $d := (1, 0^{d_P})$. We will verify the conditions in Bonnans and Shapiro (2013) Theorem 4.25. In particular, we verify the directional regularity condition in the direction $d := (1, 0^{d_P})$ for all $\alpha^* \in \mathcal{A}^{\delta,*}_{I,Lip}$. By Lemma 17, $D_{KL}(F\|F_0)$ is continuously differentiable on Π_{Lip}. Therefore, the setting in Bonnans and Shapiro (2013) Page 260 is satisfied.

For $\alpha^* = (\theta^*, F^*) \in \mathcal{A}^{\delta,*}_{I,Lip}$, by Bonnans and Shapiro (2013) Theorem 4.9, the directional regularity condition in the direction d is equivalent to:

$$0 \in \operatorname{int}\left\{\begin{pmatrix} D_{KL}(F^*\|F_0) - \delta \\ P(\alpha^*) - P_0 \end{pmatrix} + \begin{pmatrix} D_{F^*} D_{KL}(F^*\|F_0)(\Pi_{Lip} - F^*) - [0, +\infty) \\ DP(\alpha^*)(\Theta \times \Pi_{Lip} - \alpha^*) - 0^{d_P} \end{pmatrix} - \begin{pmatrix} (-\infty, 0] \\ 0^{d_P} \end{pmatrix}\right\}$$

By Assumption 9(iv), the second part is satisfied. The first part is straightforward as $-[0, +\infty) - (-\infty, 0] = \mathbb{R}$.

Assumption 9(v) is the same as Assumption (iii) of Theorem 4.25 in Bonnans and Shapiro (2013). Moreover, Assumption 9(vi) is the same as Assumption (iv) of Theorem 4.25 in Bonnans and Shapiro (2013). Therefore, Bonnans and Shapiro (2013) Theorem 4.25 shows the right differentiability of $\kappa^{Lip}(\delta, P_0)$.

The proof for $\tilde{\kappa}^{Lip}_{\mathrm{TI}}(\delta, P_0)$ is the same. □

[36] See Nutz (2021) equation 1.10.

B.6 Proof in Section 7

B.6.1 Proof of Lemma 3

Proof of Lemma 3. (i) See Rust et al. (2002) Theorem 1.

(ii)
- ($\Leftarrow$) Suppose $V^*(\omega)$ is the unique fixed point of (7).

  (Existence) Note that $s_0(\omega) := 1 - \exp(\omega - V^*(\omega))$ is a fixed point of (13).

  (Uniqueness) Suppose $s_0(\omega)$ is a fixed point of (13). Let $V(\omega) := \omega - \log(1 - s_0(\omega))$. Then, we show that $V(\omega)$ is a fixed point of (7). It suffices to show that:

  $$\beta\mathbb{E}\left[V(\omega')|\omega\right] = \log\left(\exp(V(\omega)) - \exp(\omega)\right)$$

  where the right-hand side equals $\omega + \log\left(\frac{s_0(\omega)}{1-s_0(\omega)}\right)$, and the left-hand side equals $\beta\mathbb{E}\left[\omega' - \log(1 - s_0(\omega'))|\omega\right]$. Since $s_0(\omega)$ is a fixed point of (13), we have RHS = LHS. Therefore, $V(\omega)$ is a fixed point of (7).

  We prove uniqueness by contradiction. Suppose there are two fixed points $s_0(\omega)$ and $\tilde{s}_0(\omega)$ of (13). Then, we have: $V(\omega) = \omega - \log(1 - s_0(\omega))$, $\tilde{V}(\omega) = \omega - \log(1 - \tilde{s}_0(\omega))$ are both the fixed points to (7) as shown above. Since (7) has a unique fixed point, we have $V(\omega) = \tilde{V}(\omega)$ for all ω. Therefore, we have $s_0(\omega) = \tilde{s}_0(\omega)$ for all ω. Contradiction.

- ($\Rightarrow$) Suppose (13) has a unique fixed point $s_0^*(\omega)$.

  (Existence) Note that $V(\omega) := \omega - \log(1 - s_0^*(\omega))$ is a fixed point of (7).

  (Uniqueness) Suppose $V(\omega)$ is a fixed point. Then, we show that $s_0(\omega) := 1 - \exp(\omega - V(\omega))$ is a fixed point of (13). As $V(\omega) = \omega - \log(1 - s_0(\omega))$ is a fixed point of (7), we have: $\exp(\omega - \log(1 - s_0(\omega))) = \exp(\omega) + \exp\left(\beta\mathbb{E}\left[\omega' - \log(1 - s_0(\omega'))|\omega\right]\right)$. It implies $\frac{1}{1-s_0(\omega)} = 1 + \exp\left(\beta\mathbb{E}\left[\omega' - \log(1 - s_0(\omega'))|\omega\right] - \omega\right)$. Rearranging it shows that $s_0(\omega)$ is a fixed point of (13).

  We prove uniqueness by contradiction. Suppose there are two fixed points $V_1(\omega)$ and $V_2(\omega)$ of (7). Then, we have: $s_0(\omega) = 1 - \exp(\omega - V_1(\omega))$, $\tilde{s}_0(\omega) = 1 - \exp(\omega - V_2(\omega))$ are both the fixed points to (13) as shown above. Since (13) has a unique fixed point, we have $s_0(\omega) = \tilde{s}_0(\omega)$ for all ω. Therefore, we have $V_1(\omega) = V_2(\omega)$ for all ω. Contradiction.

(iii) This is a direct consequence of the above argument.

□